

Product Switching and Young Firm Dynamics*

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June 15, 2026

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Abstract

We develop a model of product switching over the firm lifecycle in which firms learn about product-specific expertise and adjust their product portfolios accordingly. Firms collect and process information to extract signals, and their information-processing capacity improves over time. However, limited data processing ability early in the lifecycle generates noisier beliefs and distinctive product-switching behavior among young firms: young firms are more likely to retain poorly matched products and drop better-matched ones, slowing the reallocation of resources toward high-performing products. This mechanism introduces a novel within-firm margin of resource allocation through product switching that complements the standard firm selection channel and provides a framework for evaluating policies such as product search and entry subsidies. Using firm-product-level administrative data and the Management and Organizational Practices Survey (MOPS) from the U.S. manufacturing sector, we find product switching patterns across the firm lifecycle and managerial ability that are consistent with the model's predictions.

JEL Code: L11, L21, L25, L60, O31

Keywords: product add and drop, product switching, young firm dynamics, firm growth, resource allocation, aggregate productivity

*We are grateful to David Argente, Gadi Barlevy, Isaac Baley, Salome Baslandze (discussant), Andres Blanco, Johannes Boehm, Satyajit Chatterjee, Joel David, Marteen De Ridder (discussant), Jason Faberman, Alessandro Ferrari, Stefania Garetto, John Haltiwanger, Bart Hobijn, Hugo Hopenhayn, Adrien Matray, Ryan Michaels, Javier Miranda, Alex Monge-Naranjo, Sara Moreira (discussant), Jane Olmstead-Rumsey, Marta Prato, Felipe Saffie, John Shea, Shihan Shen, Stephen Redding, Richard Rogerson, Leena Rudanko, Jim Tybout, Steve Yeaple, and participants at multiple seminars and conferences, including the Federal Reserve Banks of Atlanta, Chicago, Philadelphia, Rice University, CEPR Women in Economics seminar, Boston University TPRI, Brookings Institution, U.S. Census Bureau, Kiel Institute for the World Economy, Bank of Korea, Sungkyunkwan University, the ASSA, BSE Summer Forum, CAED, CEPR Joint Workshop IMO-ESF, Midwest Macro, SED, Econometric Society World Congress Meetings for helpful comments. Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2095. (CBDRB-FY25-P2095-R12268, CBDRB-FY26-P2095-R13144) All errors are our own.

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1 Introduction

Finding an optimal set of products to produce is important for how firms allocate resources. Firms add and drop products throughout their life-cycle, and the literature finds that product switching is prevalent and has a profound impact on a firm scope and performance (Bernard et al., 2010; Argente et al., 2018). When switching products, firms often remain close to their existing ones (e.g., Bernard et al., 2010; Boehm et al., 2022). However, firms also venture into entirely new industries, and such radical product switching can yield substantial performance gains and contribute to long-term success.¹ At the same time, many firms fail to introduce new products successfully, suggesting that they may excel in certain areas while struggling in others.²

These decisions may be particularly important when firms are young, as their production scope and specialty are not yet fully established. As a result, they may need to search more intensively to find the right products to produce. These dynamics have important aggregate implications, as the growth of young firms is a key driver of aggregate productivity and economic growth (Haltiwanger, 2012; Haltiwanger et al., 2013; Decker et al., 2014, 2016; Haltiwanger et al., 2016; Foster et al., 2018).

In this paper, we study firms’ product-switching decisions over the lifecycle and their implications for firm growth and resource allocation. We develop a model in which firms learn about how well their products perform in the market—how appealing products are to consumers—through production and sales, which we term product-specific expertise. Firms use their beliefs about expertise and adjust their product portfolios over the lifecycle. Our framework highlights a novel mechanism: learning frictions during the early firm lifecycle slow the formation of accurate beliefs, resulting in systematically different production and product-switching decisions across firms in different lifecycle.

¹For example, Samsung started as a grocery trading company, but by shifting its core business to electronics and semiconductor manufacturing, it eventually evolved into the globally recognized firm as is today. Similarly, 3M began as a mining venture but became renowned by diversifying into a wide range of products for worker safety, healthcare, and consumer products.

²For instance, Dyson and Apple both attempted to enter the electric car market but ultimately failed. Likewise, McDonald’s was unsuccessful in its attempt to sell pizza, despite potential complementarities in inputs and distribution networks.

We show that this mechanism has important implications for resource allocation across product markets, creating inefficiencies in firms' product search, product-dropping, and entry and exit decisions in equilibrium. Using firm-product-level administrative data from the U.S. manufacturing sector, we provide evidence consistent with the model's key predictions.

The model builds on the learning framework of [Jovanovic \(1982\)](#) and product switching behavior of multi-product firms in [Bernard et al. \(2010\)](#). Firms are born with heterogeneous types that determine productivity, product scope, and their baseline ability to process information and extract signals; this information-processing ability improves over time through experience and data accumulation. They are also endowed with time-invariant expertise in each product market, which is initially unobserved. Firms form prior beliefs about this expertise and update them over time using realized market-clearing prices and extracted signals through a Bayesian learning process. Based on their beliefs, firms decide whether to search for a new product to add or to switch existing ones, incurring a search cost.

The model generates rich lifecycle patterns in product-switching dynamics: (i) Noisy beliefs early in the lifecycle. When firms are young, lower information-processing ability leads to noisier signals and greater reliance on the prior, resulting in systematic underestimation of high-potential products (high product-specific expertise) and overestimation of low-potential products, along with more diffuse posterior beliefs. (ii) Belief-driven patterns in production. The noisy beliefs translate into a wedge between perceived and true product potential, leading firms to produce relatively less of high-potential products and relatively more of low-potential products. (iii) Systematic patterns in product-switching behavior. As a consequence, when firms are young, they are less likely to retain high-potential products and more likely to retain low-potential products, resulting in misaligned product-portfolio decisions early in the lifecycle.

These mechanisms have novel aggregate implications for resource allocation. We show that learning frictions can generate inefficiencies along multiple extensive margins, including firm entry and exit as well as product-market entry and exit within firms.

The latter margin is new, which reallocates resources across product markets through product-portfolio adjustments by firms. The source of these inefficiencies is the interaction between learning and monopolistic competition. When making entry, exit, search, or product-dropping decisions, firms do not internalize either the benefits that additional product varieties create for consumers or the business-stealing effects imposed on competing firms. In our framework, learning creates an option value associated with future experimentation and information acquisition, which can cause excessive or insufficient firm entry and exit as well as inefficient product search and dropping decisions.

These findings point to a potential role for policies such as entry subsidies, product-search subsidies, or interventions that influence firms' product-dropping and exit decisions. The model provides a framework for quantifying the contribution of learning frictions to aggregate outcomes and for evaluating policies that operate through different extensive margins. In particular, it allows us to assess the aggregate importance of product-portfolio adjustments of firms, which is a new extensive margin, and to study the effectiveness of policies designed to influence product search, product reallocation, and product exit decisions.

Next, we use a detailed firm-product level dataset for the manufacturing sector from the U.S Census Bureau from 1982 to 2007 to test the model's predictions. In this dataset, we define a "product" as a five-digit Standard Industrial Classification (SIC) category for the pre-2002 years and a seven-digit North American Industry Classification System (NAICS) category from 2002 onward. We construct product-firm pairs in each Census year. A key contribution of our empirical approach is the estimation of firms' time-invariant product-specific expertise using firm-product fixed effects from real shipment values.³ To our knowledge, this represents a novel contribution of the paper.

When estimating time-invariant product-specific expertise, we control for time-varying

³Our approach closely follows [Bonhomme et al. \(2019\)](#), which estimates worker-firm complementarities in earnings.

compounding factors such as product life-cycle dynamics (e.g., [Argente et al., 2024](#)), customer capital accumulation, and transitory product-level shocks (e.g., fluctuations in market conditions) that may obscure the firm-specific ability. The resulting measure isolates a firm’s underlying capacity to produce and sell a given product, capturing the extent of its expertise in that product market.

Using this expertise measure, we empirically investigate how the product-switching behavior of firms vary across firms in different lifecycle or with different managerial ability measured by the accompanying data in the Census, the Management and Organizational Practices Survey (MOPS). Furthermore, we examine how these decisions affect subsequent firm growth and performance. We document a set of empirical patterns that support the model’s predictions: firms generally drop products with low product-specific expertise more frequently. However, when firms are young or with lower managerial ability, they exhibit a distinct pattern. They drop products less frequently overall, but are more likely to drop high-expertise products and less likely to drop low-expertise products. These findings are consistent with our model.

In addition, we find that product switching plays an important role in firm growth and performance. In particular, product churning—the simultaneous adding and dropping of products—is positively associated with multiple measures of firm performance, including firm growth in value added, employment, total factor productivity, and labor productivity. Furthermore, decomposing firm-level growth into the growth of continuing products, reallocation across continuing products, and product switching, we find that product switching accounts for a substantial share of firm growth, particularly among young firms. Combined with the observed product-switching patterns of young firms, these results suggest that firms face greater challenges in identifying and retaining the right products when they are young. Their limited ability to shed low-potential products and adopt high-potential ones early in the lifecycle prevents them from operating in product markets where they have strong expertise and constrains performance improvements.

Literature Review. Our paper is closely related to a line of recent studies on product switching within firms. [Bernard et al. \(2010\)](#) is the closest study to our paper. They use the same dataset as ours (the Census of Manufactures) and study the extent of product switching within firms for the U.S. manufacturing sector. [Broda and Weinstein \(2010\)](#) document the patterns of product entry and exit in consumer goods sectors from the Nielsen Homescan database. [Bernard and Okubo \(2016\)](#) study the role of product adding and dropping within Japanese manufacturing firms over the business cycle. [Argente et al. \(2018\)](#) use the Nielsen Retail Measurement Services (RMS) scanner data and assess the magnitude of product creation and destruction and product reallocation during and after the Great Recession. [Berlingieri et al. \(2024\)](#) use data on French manufacturing firms and document that rapid, simultaneous additions of many new products play a central role in firm innovation and growth. Other studies show that trade liberalization and competition lead firms to focus on their core products and competence ([Bernard et al., 2011](#); [Eckel and Neary, 2010](#); [Mayer et al., 2014](#)). [Boehm et al. \(2022\)](#) find that firms choose products with common-input capabilities. This paper contributes to this literature by offering a new direct measure of product-specific expertise of firms and establishing novel facts about the product match-quality ladder within firms and its evolution over the firm life-cycle.

This paper also adds to the vast literature on potential factors or frictions that affect the post-entry dynamics and growth of young firms. One strand of studies emphasizes the importance of financing constraints and the role of collateral values for successful young firms ([Evans and Jovanovic, 1989](#); [Holtz-Eakin et al., 1994](#); [Cooley and Quadrini, 2001](#); [Hurst and Lusardi, 2004](#); [Kerr and Nanda, 2009](#); [Robb and Robinson, 2014](#); [Schmalz et al., 2017](#); [Davis and Haltiwanger, 2019](#)). On the other hand, [Foster et al. \(2016\)](#) analyze the process of accumulating demand can create friction for new businesses to grow high. [Akcigit and Ates \(2019\)](#) and [Jo and Kim \(2024\)](#) present barriers to knowledge spillovers as a source to create frictions dampening firm entry and the rapid-growth young firm activities. Furthermore, [Kim \(2025\)](#) demonstrates that uncertain job prospects can pose difficulties to young firms in attracting workers

properly and negatively affect firm entry and the growth of high-potential young firms. Along this line, this paper provides a unique channel of product switching to account for young firm growth. In particular, we document new data evidence suggesting that young firms have more difficulties in finding the right product match, and this can give a negative impact on their growth path.

The rest of the paper proceeds as follows. Section 2 presents a theoretical framework of product switching with learning. Section 3 describes data sources and main measures used in our analysis. Section 4 presents our main empirical findings on product dropping and adding activities over the firm lifecycle, the relationship between product dropping and the properties of added products, and the impact of product switching on firm growth, consistent with the model’s predictions. Section 4.5 discusses additional robustness tests for the main results. Section 5 calibrates the model and conducts counterfactual exercises (in progress). Finally, Section 6 concludes with a discussion on remaining future work.

2 Theoretical Framework

Firms in the early lifecycle may lack insights into market dynamics and may not fully understand their own potential (e.g., Jovanovic, 1982; Kim, 2025). As a result, they may struggle to assess which products are aligned with their expertise and better attract consumer demand in the market, in contrast to mature firms with better experience and information.

This section describes a model of product switching over the firm lifecycle with learning about product markets. The model builds on the learning process in Jovanovic (1982), which varies along the firm lifecycle given the nascency of young firms, and product switching behavior of multi-product firms in Bernard et al. (2010). In addition, it incorporates heterogeneity in firms’ information-processing ability, motivated by the data economy literature (e.g., Farboodi et al., 2019; Farboodi and Veldkamp, 2021; Baley and Veldkamp, 2025).

2.1 Environment

The model consists of a finite set of products $p \in \{1, 2, \dots, N\}$ and a continuum of firms (i) operating in each product market. Product markets are monopolistically competitive, with firms producing differentiated varieties within each product. Time is discrete, denoted by t .

2.2 Consumers

There is a representative household with labor endowment $\bar{L}$ that supplies labor inelastically and owns all firms. The household chooses consumption to maximize

$$\sum_{t=0}^{\infty} \beta^t \log \bar{C}_t$$

subject to the period budget constraint

$$\sum_{p=1}^N \int_{i \in \Omega_{pt}} p_{ipt} c_{ipt} di + w_t \mathcal{F}^e \leq w_t L + \Pi_t,$$

where Ω_{pt} denotes the set of firms operating in product market p , p_{ipt} is the price of good produced by firm i in product p , c_{ipt} is the consumption of it, $w_t \mathcal{F}^e$ is the total expenditure used to finance entry costs for potential startups in each period, w_t is the wage, and Π_t is aggregate profits rebated by firms.

The aggregate consumption index $\bar{C}_t$ is determined by a constant elasticity of substitution (CES) aggregator over a finite set of products $p \in 1, 2, \dots, N$ available in the economy:

$$\bar{C}_t = \left(\sum_{p=1}^N C_{pt}^{\frac{\nu-1}{\nu}} \right)^{\frac{\nu}{\nu-1}},$$

where the composite goods in product market p , C_{pt} , is given by

$$C_{pt} = \left[\int_{i \in \Omega_p} \theta_{ipt}^{\frac{1}{\eta}} c_{ipt}^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}},$$

which is a CES aggregator over differentiated goods produced by firms operating in market p . The term θ_{ipt} is a product-specific demand shifter that captures how well firm i 's product performs in the market.⁴ The parameters ν and η denote the constant elasticities of substitution across products and across goods within each product market, respectively, with $1 < \nu < \eta$.

The household minimizes the cost of attaining $\bar{C}_t$ given the budget constraint, which yields the following demand function for each good:⁵

$$c_{ipt} = P_{pt}^{\eta} C_{pt} \theta_{ipt} p_{ipt}^{-\eta}. \quad (2.1)$$

The aggregate price index associated with the composite good $\bar{C}_t$ is

$$\bar{P}_t = \left(\sum_{p=1}^N P_{pt}^{1-\nu} \right)^{\frac{1}{1-\nu}}, \quad (2.2)$$

where the product-level price index is given by

$$P_{pt} = \left(\int_{i \in \Omega_p} \theta_{ipt} p_{ipt}^{1-\eta} di \right)^{\frac{1}{1-\eta}}. \quad (2.3)$$

2.3 Firms

Firms are born with a fundamental type φ_i , which determines productivity $z_i \equiv z(\varphi_i)$ in production, capacity $n_i \equiv n(\varphi_i)$, which is the maximum number of product lines that

⁴ θ_{ipt} captures firm-product-specific fundamentals that determine market performance, such as how appealing the product is to consumers and how strongly it aligns with demand conditions. We refer to this as the firm's product-specific expertise.

⁵The corresponding market-clearing price is $p_{ipt} = P_{pt} C_{pt}^{\frac{1}{\eta}} \theta_{ipt}^{\frac{1}{\eta}} c_{ipt}^{-\frac{1}{\eta}}$.

the firm can operate at a time and determines the scope of the firm, and managerial practices $m_i \equiv m(\varphi_i)$. The mappings satisfy $z : \mathbb{R} \rightarrow \mathbb{R}$ with $z'(\cdot) > 0$, $n : \mathbb{R} \rightarrow \mathbb{N}$ with $N'(\cdot) > 0$, and $m : \mathbb{R} \rightarrow \mathbb{R}$ with $m'(\cdot) > 0$. In each period, firms decide for each product line whether to add or drop a product and choose quantities for all operating products.

2.3.1 Firm-Product Match and Learning

In addition to their fundamental type, firms (i) are born with time-invariant, product-specific expertise θ_{ip} for each product p . θ_{ip} is drawn from a log-normal distribution, $\log \theta_{ip} \sim N(\theta_0, \sigma_0^2)$. A higher value of θ_{ip} indicates a better match between the firm and the product, reflecting greater expertise in operating that product, and thus leading to higher profitability. We assume that this expertise is embodied in product quality and enters the demand function.⁶

Firms do not observe θ_{ip} directly. Instead, in each period they observe a noisy signal inferred from the market-clearing price induced by realized demand:

$$\log \theta_{ipt} = \log \theta_{ip} + \varepsilon_{ipt},$$

where $\varepsilon_{ipt} \sim N(0, \sigma_\varepsilon^2)$ is an i.i.d. shock that follows a normal distribution. We assume that learning follows Bayesian updating based on $\log \theta_{ipt}$ and an additional signal $\log s_{ipt}$ obtained each period (Farboodi et al., 2019; Farboodi and Veldkamp, 2021; Baley and Veldkamp, 2025). The signal is extracted using the firm's information-processing ability ψ_{it} :

$$\log s_{ipt} = \log \theta_{ip} + v_{ipt}, \quad v_{ipt} \sim N(0, \frac{\sigma_\varepsilon^2}{\psi_{it}}). \quad (2.4)$$

A firm's ability to process observable information depends on its managerial practices m_{it} (a function of φ_i and, potentially, training), its experience (captured by firm age, a_{it}), and the size of its accumulated data D_{it} (which depends on a_{it} , product scope n_i , and potentially other data sources). We summarize these factors by $\psi_{it} \equiv$

⁶Alternatively, it can be modeled on the production side as product-level productivity.

$\psi(a_{it}, m_{it}, D_{it}; \varphi_i)$, with $\partial\psi_{it}/\partial x \geq 0, \forall x \in \{a_{it}, m_{it}, D_{it}, \varphi_i\}$.⁷ This ability increases the precision of the signals firms draw. If managerial training and other data sources are exogenously given and fixed, then firm age (a_{it}) fully determines the evolution of ψ_{it} .⁸

Given this structure, learning proceeds as follows. Firm i enters product market p with an initial belief about θ_{ip} , given by the prior $\log \theta_{ip} \sim N(\theta_0, \sigma_0^2)$. After production and sales, the firm observes realized outcome and updates its beliefs each period. In period t , conditional on having operated in product market p for τ periods (firm-product tenure) and observing the contemporaneous outcome θ_{ipt} , its information set is

$$\mathcal{I}_{ipt} \equiv \{\theta_{ip\tilde{t}}, s_{ip\tilde{t}}\}_{\tilde{t}=t-\tau+1}^t. \quad (2.5)$$

This yields the posterior belief as follows:

$$\hat{\theta}_{ipt} = \frac{\left(\frac{\theta_0}{\sigma_0^2} + \frac{\tau}{\sigma_\varepsilon^2} \frac{\sum_{\tilde{t}=t-\tau+1}^t \log \theta_{ip\tilde{t}}}{\tau} + \frac{\sum_{\tilde{t}=t-\tau+1}^t \psi_{i\tilde{t}} \log s_{ip\tilde{t}}}{\sigma_\varepsilon^2}\right)}{\left(\frac{1}{\sigma_0^2} + \frac{\tau}{\sigma_\varepsilon^2} + \frac{\sum_{\tilde{t}=t-\tau+1}^t \psi_{i\tilde{t}}}{\sigma_\varepsilon^2}\right)} \quad (2.6)$$

$$\hat{\sigma}_{ipt}^2 = \frac{1}{\left(\frac{1}{\sigma_0^2} + \frac{\tau}{\sigma_\varepsilon^2} + \frac{\sum_{\tilde{t}=t-\tau+1}^t \psi_{i\tilde{t}}}{\sigma_\varepsilon^2}\right)}, \quad (2.7)$$

where $\hat{\theta}_{ipt}$ and $\hat{\sigma}_{ipt}^2$ is the mean and variance of the firm's posterior belief about $\log \theta_{ip}$, respectively.

Importantly, the posterior mean is a weighted average of the prior θ_0 , the average observed outcome $\frac{1}{\tau} \sum_{\tilde{t}=t-\tau+1}^t \log \theta_{ip\tilde{t}}$, and signals $\{\log s_{ip\tilde{t}}\}_{\tilde{t}=t-\tau+1}^t$, with weights determined by tenure τ and the firm's time-varying information-processing ability $\{\psi_{i\tilde{t}}\}_{\tilde{t}=t-\tau+1}^t$. This structure allows us to identify multiple sources of learning and to disentangle the role of the firm's information-processing ability (which evolves with firm age) from the effect of product-specific tenure. Given this, we can characterize the posterior mean and variance of θ_{ip} , as well as beliefs about future realizations θ_{ipt+1} .

⁷The cumulative number of products dropped can serve as an additional proxy for firm experience. We abstract from this variable because it introduces dependencies across product lines, which substantially reduce the tractability of the model.

⁸Then, the state variables that need to be tracked reduce to $(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \varphi_i)$.

Note that as the posterior variance is fully determined by product–firm tenure, firm age, information-processing ability, and firm type, posterior beliefs are sufficiently characterized by the set of $(\hat{\theta}_{ipt}, \tau; a_{it}, \psi_{it}, \varphi_i)$. All firms start with a common initial prior $(\hat{\theta}_{ip}^0, 0; a_{it}, \psi_{it}, \varphi_i)$, corresponding to $(\hat{\theta}_{ip}^0, \hat{\sigma}_{ip}^0) \equiv (\theta_0, \sigma_0)$. The following lemma presents how posterior beliefs about product-specific expertise θ_{ip} vary with firms' information-processing ability over the lifecycle, even conditional on the same set of observed outcomes $\{\theta_{ipt}\}_{t=t-\tau+1}^t$.

Lemma 1 (Belief Formation). *Holding all else equal, lower information-processing ability ψ_{it} underestimates $\hat{\theta}_{ipt}$ for high- θ_{ip} products and overestimates $\hat{\theta}_{ipt}$ for low- θ_{ip} products, and obtains higher posterior variance.*

Proof. The first derivative of $\hat{\theta}_{ipt}$ with respect to ψ_{it} is given by

$$\frac{\partial \hat{\theta}_{ipt}}{\partial \psi_{it}} = \frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2} \left[\log s_{ipt} \left(\frac{1}{\sigma_0^2} + \frac{\tau}{\sigma_\varepsilon^2} + \frac{\sum_{\bar{t}} \psi_{i\bar{t}}}{\sigma_\varepsilon^2} \right) - \left(\frac{\theta_0}{\sigma_0^2} + \frac{\sum_{\bar{t}} \log \theta_{ip\bar{t}}}{\sigma_\varepsilon^2} + \frac{\sum_{\bar{t}} \psi_{i\bar{t}} \log s_{ip\bar{t}}}{\sigma_\varepsilon^2} \right) \right].$$

Since $\log \theta_{ipt} = \log \theta_{ip} + \varepsilon_{ipt}$, where $\varepsilon_{ipt} \sim N(0, \sigma_\varepsilon^2)$, and $\log s_{ipt} = \log \theta_{ip} + v_{ipt}$, where $v_{ipt} \sim N(0, \frac{\sigma_\varepsilon^2}{\psi_{it}})$, taking expectations with respect to the noise terms yields

$$\mathbb{E}_{\varepsilon, v} \left(\frac{\partial \hat{\theta}_{ipt}}{\partial \psi_{it}} \right) = \frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2 \sigma_0^2} (\log \theta_{ip} - \theta_0).$$

Therefore, on average, when $\log \theta_{ip} > \theta_0$, a higher ψ_{it} raises the posterior mean, implying that firms with lower ψ_{it} underestimate $\hat{\theta}_{ipt}$. Conversely, when $\log \theta_{ip} < \theta_0$, a higher ψ_{it} lowers the posterior mean, so firms with lower ψ_{it} overestimate $\hat{\theta}_{ipt}$. Moreover,

$$\frac{\partial \hat{\sigma}_{ipt}^2}{\partial \psi_{it}} = -\frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2} < 0.$$

Therefore, firms with lower information-processing ability have a higher posterior variance. $\square$

Since ψ_{it} is closely related to firm age a_{it} , Lemma 1 implies that firms exhibit system-

atically different posterior belief formation when they are young. Higher information-processing ability makes their signals more precise and informative. In addition, it increases the weight associated with these signals when forming posterior beliefs. As ψ_{it} increases with age, this implies that older firms place higher weights on more precise signals, which makes their estimates $\hat{\theta}_{ipt}$ closer to actual value θ_{ip} . This leads to systematically different product-switching decisions along the firm lifecycle.

Lastly, the transition probability of $\hat{\theta}_{ipt}$ can be defined in the following way. Under our functional-form assumption, ψ_{it} is fully determined by (a_{it}, φ_i) , and conditional on continuing production of product p , tenure evolves deterministically.⁹ At time t , conditional on current information, $\log \theta_{ip} \mid (\hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i) \sim N\left(\hat{\theta}_{ipt}, \hat{\sigma}_{ipt}^2\right)$, where $\hat{\sigma}_{ipt}^2$ is determined by τ_{ipt} , firm age, and firm type, as described by (2.7). In period $t+1$, the firm observes $\log \theta_{ipt+1} = \log \theta_{ip} + \varepsilon_{ipt+1}$, and $\log s_{ipt+1} = \log \theta_{ip} + v_{ipt+1}$. Bayesian updating then implies $\hat{\sigma}_{ipt+1}^2 = \left(\frac{1}{\hat{\sigma}_{ipt}^2} + \frac{1+\psi_{it+1}}{\sigma_\varepsilon^2}\right)^{-1}$, and $\hat{\theta}_{ipt+1} = \hat{\sigma}_{ipt+1}^2 \left(\frac{\hat{\theta}_{ipt}}{\hat{\sigma}_{ipt}^2} + \frac{\log \theta_{ipt+1}}{\sigma_\varepsilon^2} + \frac{\psi_{it+1} \log s_{ipt+1}}{\sigma_\varepsilon^2}\right)$.

Substituting for $\log \theta_{ipt+1}$ and $\log s_{ipt+1}$ yields

$$\hat{\theta}_{ipt+1} = \underbrace{\hat{\sigma}_{ipt+1}^2 \left(\frac{\hat{\theta}_{ipt}}{\hat{\sigma}_{ipt}^2} + \frac{(1 + \psi_{it+1}) \log \theta_{ip}}{\sigma_\varepsilon^2} \right)}_{\text{depends on } \log \theta_{ip}} + \underbrace{\hat{\sigma}_{ipt+1}^2 \frac{\varepsilon_{ipt+1} + \psi_{it+1} v_{ipt+1}}{\sigma_\varepsilon^2}}_{\text{noise}}.$$

Integrating out $\log \theta_{ip}$, ε_{ipt+1} , and v_{ipt+1} conditional on the current posterior yields $\hat{\theta}_{ipt+1} \mid (\hat{\theta}_{ipt}, \tau_{ipt}) \sim N\left(\hat{\theta}_{ipt}, \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2\right)$, with $\tau_{ip,t+1} = \tau_{ipt} + 1$.¹⁰ Therefore, the

⁹For simplicity, we assume that m_{it} is fixed as a function of φ_i and D_{it} is fully characterized by the accumulated set of information up to t , determined by the dimension of $n_i \times a_{it}$.

¹⁰The posterior mean is normally distributed because it is a linear function of Gaussian signals and a Gaussian prior, implying that belief updating preserves normality. The variance term follows from the law of total variance:

$$\begin{aligned} \text{Var}(\log \theta_{ip} \mid \hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i) &= \mathbb{E} \left[\text{Var}(\log \theta_{ip} \mid \hat{\theta}_{ipt+1}, \tau_{ipt+1}, a_{it+1}, \varphi_i) \mid \hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i \right] \\ &\quad + \text{Var} \left(\mathbb{E}[\log \theta_{ip} \mid \hat{\theta}_{ipt+1}, \tau_{ipt+1}, a_{it+1}, \varphi_i] \mid \hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i \right), \end{aligned}$$

where $\text{Var}(\log \theta_{ip} \mid \hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i) = \hat{\sigma}_{ipt}^2$, $\text{Var}(\log \theta_{ip} \mid \hat{\theta}_{ipt+1}, \tau_{ipt+1}, a_{it+1}, \varphi_i) = \hat{\sigma}_{ipt+1}^2$, and $\mathbb{E}[\log \theta_{ip} \mid \hat{\theta}_{ipt+1}, \tau_{ipt+1}, a_{it+1}, \varphi_i] = \hat{\theta}_{ipt+1}$.

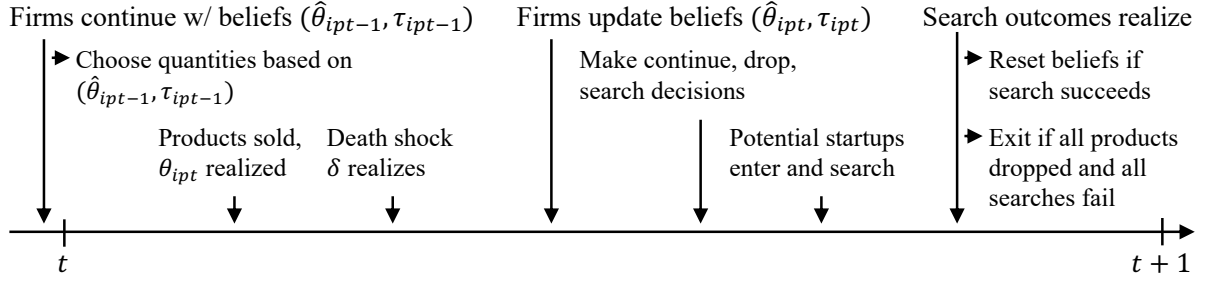


Figure 1: Timeline of the model

transition probability $g_{it}(\cdot \mid \hat{\theta}_{ipt}, \tau_{ipt}) \equiv g(\cdot \mid \hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i)$ takes the form

$$g_{it}(\hat{\theta}_{ipt+1}, \tau_{ipt+1} \mid \hat{\theta}_{ipt}, \tau_{ipt}) = \phi\left(\hat{\theta}_{ipt+1}; \hat{\theta}_{ipt}, \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2\right) \mathbf{1}\{\tau_{ipt+1} = \tau_{ipt} + 1\}, \quad (2.8)$$

where $\phi(\hat{\theta}_{ipt+1}; \hat{\theta}_{ipt}, \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2) = \frac{1}{\sqrt{2\pi(\hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2)}} \exp\left\{-\frac{(\hat{\theta}_{ipt+1} - \hat{\theta}_{ipt})^2}{2(\hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2)}\right\}$. The variance term $\hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2$ corresponds to the reduction in posterior uncertainty between periods t and $t+1$.

This implies that $\hat{\theta}_{ip,t+1}$ is a martingale with respect to the information filtration. Therefore, product p produced by firm i can be fully characterized by the state space $(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \varphi_i)$, which defines the product-line value function studied below.¹¹

2.3.2 Timing of the Model

The timing of the model is illustrated in figure 1. At the beginning of period t , firms enter with initial posterior beliefs about θ_{ip} and predetermined quantities, both based on $\mathcal{I}_{ipt-1}$, prior to the realization of θ_{ipt} .¹² Demand and sales are then realized given θ_{ipt} , after which the death shock δ is realized. Surviving firms update their posterior beliefs based on realized outcomes and decide whether to continue operating or drop

¹¹We express posterior beliefs using the state vector $(\hat{\theta}, \tau; a, \varphi)$ to highlight a key distinction in our model: firm age and firm-product tenure play separate roles in the learning process. In particular, firm age affects information processing ability independently of product-line tenure. This formulation also simplifies computation, as $\hat{\theta}$ is the only continuous endogenous state variable, while τ and a evolve discretely over time and φ is time invariant.

¹²For entrant firms, the initial posterior coincides with the initial prior $(\hat{\theta}_{ip}^0, \hat{\sigma}_{ip}^0)$.

the product, and whether to switch to a new product via search.

Potential startups also conduct searches to enter product markets. Search outcomes are realized at the end of the period. A firm exits the economy if it drops all of its products and fails all of its searches. Firms that continue into the next period choose quantities based on updated beliefs, where firms that switch and successfully enter a new product market reset their beliefs to the initial prior $(\hat{\theta}_{ipt}^0, \hat{\sigma}_{ipt}^0)$. Accordingly, we define firm decisions and value function after θ_{ipt} is realized and beliefs are updated, when the search decision is made.

2.3.3 Production

The firm produces each product using linear production technology with labor as the only input, $y_{ipt} = z_i l_{ipt}$. Firms choose quantity based on their posterior beliefs about θ_{ip} before observing the contemporaneous realization θ_{ipt} in each period. Production incurs an operating fixed cost f^o , measured in units of labor, for each product.

For continuing products at $t + 1$, the firm chooses output y_{ipt+1} to maximize expected per-period profit $\mathbb{E}_{\theta_{ipt+1}} \pi_{it+1}(\hat{\theta}_{ipt}, \tau_{ipt}) \equiv \mathbb{E}_{\theta_{ipt+1}} \pi(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i)$ with $z_i = z(\varphi_i)$, where expectations are taken over θ_{ipt+1} conditional on the firm's posterior beliefs $(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i)$:

$$\begin{aligned} \mathbb{E}_{\theta_{ipt+1}} \pi_{it+1}(\hat{\theta}_{ipt}, \tau_{ipt}) &\equiv \max_{y_{ipt+1}} \mathbb{E}_{\theta_{ipt+1}} \left[p_{ipt+1}(\theta_{ipt+1}) | (\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i) \right] y_{ipt+1} \\ &\quad - \frac{w_{t+1}}{z_i} y_{ipt+1} - w_{t+1} f^o \end{aligned} \quad (2.9)$$

subject to the demand function (2.1). The optimal quantity choice is

$$y_{ipt+1} = \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt+1}^\eta C_{pt+1} \left[\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) \right]^\eta z_i^\eta w_{t+1}^{-\eta}, \quad (2.10)$$

where $\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) \equiv \mathcal{B}(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i) \equiv \mathbb{E}_{\theta_{ipt+1}} \left[\theta_{ipt+1}^{1/\eta} | (\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i) \right]$ de-

notes the posterior mean of $\theta_{ipt+1}^{1/\eta}$ given $(\hat{\theta}_{ipt}, \tau_{ipt})$:

$$\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) = \exp \left\{ \frac{\hat{\theta}_{ipt}}{\eta} + \frac{1}{2} \left(\frac{\hat{\sigma}_{ipt}^2 + \sigma_\varepsilon^2}{\eta^2} \right) \right\}. \quad (2.11)$$

Substituting this optimal quantity into (2.1), the market-clearing price, conditional on the realized θ_{ipt} is

$$p_{ipt+1} = \frac{\eta}{\eta - 1} \frac{w_{t+1}}{z_i} \frac{\theta_{ipt+1}^{1/\eta}}{\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt})}. \quad (2.12)$$

Thus, prior to observing θ_{ipt+1} , the firm's expected price is the standard constant markup over marginal cost:

$$\mathbb{E}_{\theta_{ipt+1}} \left[p_{ipt+1} | (\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i) \right] = \frac{\eta}{\eta - 1} \frac{w_{t+1}}{z_i}. \quad (2.13)$$

Substituting (2.13) into (2.9), the optimal expected per-period profit in product market p is given by

$$\begin{aligned} \bar{\pi}_{it+1}(\hat{\theta}_{ipt}, \tau_{ipt}) &\equiv \mathbb{E}_{\theta_{ipt+1}} \pi_{it+1}(\hat{\theta}_{ipt}, \tau_{ipt}) \\ &= \frac{(\eta - 1)^{\eta-1}}{\eta^\eta} \left(\frac{z_i}{w_{t+1}} \right)^{\eta-1} P_{pt+1}^\eta C_{pt+1} \left[\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) \right]^\eta - w_{t+1} f^o. \end{aligned} \quad (2.14)$$

2.3.4 Product Switching and Exit

Each firm i is endowed with a fixed number of product lines $k \in \{1, 2, \dots, n_i\}$, which represent capacity slots that can be allocated to different products over time. At any point in time, each product line k is assigned to at most one product $p \in \{1, \dots, N\} \cup \emptyset$, where $\emptyset$ denotes a vacant line. The mapping from product lines to products is endogenous and evolves over time, where the destination product market is determined randomly. In particular, a firm can reassign a product line to a different product through switching.¹³ Product lines are therefore the units of capacity and decision-making,

¹³For example, a line initially assigned to product $p = 1$ can be reassigned to product $p = 3$ in a subsequent period. This distinction implies that product lines are the units of capacity and decision-making, while product indices represent the set of markets in which firms can operate.

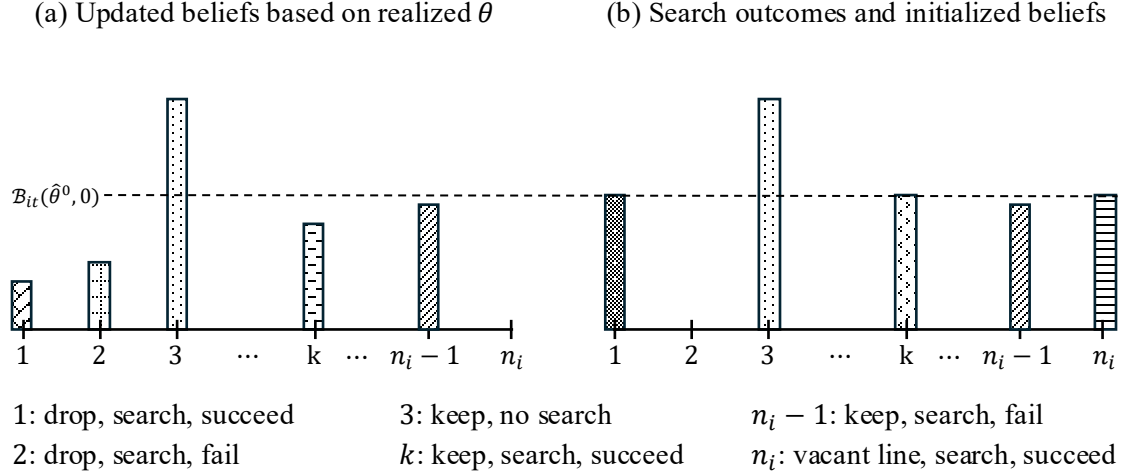


Figure 2: Evolution of product lines for firm i with scope $n_i = n(\varphi_i)$

while product indices represent the set of markets in which firms can operate.

The value of a product line depends on the firm's posterior beliefs about product-specific expertise. For each product line, the firm can initiate a search for a new product by paying a search cost f^s , measured in units of labor. Given the operating and search fixed costs, firms face two endogenous extensive-margin decisions for each product line: whether to search for a new product and whether to continue operating the existing product. These decisions give rise to endogenous search and product-drop thresholds. If the continuation value of a product line is sufficiently low, the firm either searches for a replacement product or drops the existing product and searches. If a product line is vacant and the firm does not exit, the firm searches to fill the vacant line. The firm exits if all product lines are dropped and all searches fail, or exogenously with probability δ . We formally characterize these decisions and thresholds in the next subsection.

Figure 2 illustrates possible outcomes for firm i with product scope $n_i = n(\varphi_i)$ before and after product-switching decisions. Panel (a) shows initial $B_{it}(\hat{\theta}, \tau)$ for each product line, represented by the height of each bar, and Panel (b) shows the resulting outcomes. Each product line is occupied by a distinct product, indicated by different patterns.

Products in lines 1 and 2 have $B_{it}(\hat{\theta}, \tau)$ below the drop thresholds, so the firm drops them and initiates searches. The product in line 3 has $B_{it}(\hat{\theta}, \tau)$ above the search thresh-

old, so the firm continues without searching. Products in lines k and $n_i - 1$ have $\mathcal{B}_{it}(\hat{\theta}, \tau)$ between the drop threshold and the search threshold; the firm keeps these products but initiates searches. Line n_i is vacant, so the firm searches to fill it.

Searches succeed for lines 1, k , and n_i , but fail for lines 2 and $n_i - 1$. As a result, the posterior beliefs in lines 1, k , and n_i are reset to the initial prior, $\mathcal{B}_{it}(\hat{\theta}^0, 0)$. Line 2 becomes vacant, while line n_i is filled with a new product. An additional case arises when a product line starts vacant and remains vacant due to search failure.

2.3.5 Value Function of Product Lines

Since product-switching decisions are made at the product-line level and some lines may be vacant, we formulate firm decisions and value functions at the product-line level. However, the endogenous exit decision depends on the composition of the firm's portfolio of posterior beliefs about θ_{ip} , so the product-line value function generally depends on the firm-level product portfolio. In the next section, we show that, under a plausible assumption, this linkage can be broken, allowing product-line decisions to be analyzed independently of the portfolio.

The product-line problem can be decomposed into three continuation values: no search, search while retaining the existing product, and search after dropping the product. Let $V_{it}^{(\cdot)}(\hat{\theta}_{ipkt}, \tau_{ipkt}) \equiv V^{(\cdot)}(\hat{\theta}_{ipkt}, \tau_{ipkt}; a_{it}, \psi_{it}, \varphi_i)$ for notational simplicity. The value function for product line k is given by

$$V_{it}(\hat{\theta}_{ipkt}, \tau_{ipkt}) = \max\{V_{it}^{ns}(\hat{\theta}_{ipkt}, \tau_{ipkt}), V_{it}^{cs}(\hat{\theta}_{ipkt}, \tau_{ipkt}), V_{it}^{ds}(\emptyset)\}, \quad (2.15)$$

where $V_{it}^{ns}(\hat{\theta}_{ipkt}, \tau_{ipkt})$ is the value without search, $V_{it}^{cs}(\hat{\theta}_{ipkt}, \tau_{ipkt})$ is the value with search while retaining the existing product, and $V_{it}^{ds}(\emptyset)$ is the value with search after dropping the product, which encompasses firm exit. The corresponding search and drop thresholds are denoted by $\mathcal{B}_{it}^{cs}(\tau_{ipkt})$ and $\mathcal{B}_{it}^{ds}(\tau_{ipkt})$, respectively, and are formally characterize

in Proposition 1.¹⁴ Given these thresholds, we can characterize the value function in each case.

(a) No search ($\mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) \geq \mathcal{B}_{it}^{cs}(\tau_{ip_k t})$): the firm does not conduct search in the current period. However, as $\theta_{ip_k t}$ evolves and posterior beliefs are updated, the firm may initiate search in subsequent periods.

$$V_{it}^{ns}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) = \beta \left[\bar{\pi}_{it+1}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) + (1 - \delta) \mathbb{E}_{ip_k t} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) + \delta V_{it+1}^{\delta} \right], \quad (2.16)$$

where $\mathbb{E}_{ip_k t} \equiv \mathbb{E}_{(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) | (\hat{\theta}_{ip_k t}, \tau_{ip_k t})}$, and

$$\begin{aligned} \mathbb{E}_{ip_k t} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) &= \int_{\mathcal{R}_{ip_k t+1}^{ns}} V_{it+1}^{ns}(\hat{\theta}', \tau') g_{it}(\hat{\theta}', \tau' | \hat{\theta}_{ip_k t}, \tau_{ip_k t}) d\hat{\theta}' \\ &+ \int_{\mathcal{R}_{ip_k t+1}^{cs}} V_{it+1}^{cs}(\hat{\theta}', \tau') g_{it}(\hat{\theta}', \tau' | \hat{\theta}_{ip_k t}, \tau_{ip_k t}) d\hat{\theta}' + \int_{\mathcal{R}_{ip_k t+1}^{ds}} V_{it+1}^{ds}(\emptyset) g_{it}(\hat{\theta}', \tau' | \hat{\theta}_{ip_k t}, \tau_{ip_k t}) d\hat{\theta}', \end{aligned}$$

is the expected continuation value of product line k in the absence of product switching, with posterior beliefs $(\hat{\theta}_{ip_k t}, \tau_{ip_k t})$ evolving over time. Here, $\tau' = \tau_{ip_k t} + 1$, and $g_{it}(\cdot | \hat{\theta}_{ip_k t}, \tau_{ip_k t})$ is the transition probability of posterior beliefs defined in (2.8). The regions are defined by decision rules in posterior space: $\mathcal{R}_{ip_k t+1}^{ns} = \{(\hat{\theta}', \tau'; a', \varphi) : \mathcal{B}(\hat{\theta}', \tau'; a', \varphi) \geq \mathcal{B}^{cs}(\tau'; a', \varphi)\}$, $\mathcal{R}_{ip_k t+1}^{cs} = \{(\hat{\theta}', \tau'; a', \varphi) : \mathcal{B}^{ds}(\tau'; a', \varphi) \leq \mathcal{B}(\hat{\theta}', \tau'; a', \varphi) < \mathcal{B}^{cs}(\tau'; a', \varphi)\}$, and $\mathcal{R}_{ip_k t+1}^{ds} = \{(\hat{\theta}', \tau'; a', \varphi) : \mathcal{B}(\hat{\theta}', \tau'; a', \varphi) < \mathcal{B}^{ds}(\tau'; a', \varphi)\}$. The third term on the RHS includes the firm's endogenous exit case. V_{it+1}^{δ} denotes the firm's exit value per product line under exogenous exit. We assume $V_{it+1}^{\delta} = V_{it+1}^{ds}(\emptyset)$. We discuss this in detail below.

(b) Continue/Switch with search ($\mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) < \mathcal{B}_{it}^{cs}(\tau_{ip_k t})$): In this region, the firm searches for a new product to replace its current product. The firm continues operating the current product but replaces it with a newly obtained product at the end

¹⁴We limit our analysis to the case in which $V_{it}^{ds}(\emptyset) > 0$, implying that firms always search to fill vacant product lines. The condition is characterized in (2.18). The condition under which $\mathcal{B}_{it}^{cs}(\tau_{ip_k t}) > \mathcal{B}_{it}^{ds}(\tau_{ip_k t})$ holds is characterized in (2.21).

of the period if the search succeeds. The value of this case is

$$\begin{aligned}
V_{it}^{cs}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) = & \\
& -w_t f^s + x\beta \left[\bar{\pi}_{it+1}(\hat{\theta}_{ip'_k t}, \tau_{ip'_k t}^0) + (1-\delta)\mathbb{E}_{ip'_k t}^0 V_{it+1}(\hat{\theta}_{ip'_k t+1}, \tau_{ip'_k t+1}) + \delta V_{it+1}^\delta \right] \\
& + (1-x)\beta \left[\bar{\pi}_{it+1}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) + (1-\delta)\mathbb{E}_{ip_k t} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) + \delta V_{it+1}^\delta \right], \quad (2.17)
\end{aligned}$$

where $\mathbb{E}_{ip'_k t}^0 \equiv \mathbb{E}_{(\hat{\theta}_{ip'_k t+1}, \tau_{ip'_k t+1}) | (\hat{\theta}_{ip'_k t}, \tau_{ip'_k t}^0)}$. When search succeeds with probability x and the product is switched, the firm's expertise in the new product market p' for product line k , $\theta_{ip'_k}$, is drawn from the initial prior distribution. Accordingly, the firm's posterior beliefs are reset to the initial posterior $(\hat{\theta}_{ip'_k t}, \tau_{ip'_k t}^0)$, where $\tau_{ip'_k t}^0 = 0$.

Search operates as follows. For each firm, define the exclusion set as the union of currently operated products and the most recent destination product associated with each vacant line, which has cardinality $n(\varphi_i)$. Conditional on search, each product line draws a destination product from the common feasible set of size $N - n(\varphi_i)$, with draws coordinated without replacement across product lines. The search then succeeds with probability x , in which case the destination product is adopted and replaces the existing product. For a vacant line, if search fails, the associated product is updated to the most recent destination product drawn but not adopted. Under uniform draws and symmetric product markets, expectations over product draws collapse and therefore do not appear explicitly in the value function. When search fails with probability $1 - x$, the value function coincides with the no-search case, as the firm continues to operate the existing product.

(c) drop/Switch with search ($\mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) < \mathcal{B}_{it}^{ds}(\tau_{ip_k t})$): In this case, the firm drops the product regardless of the search outcome and fills the vacant line only if search is successful; otherwise, the line remains vacant. The value of this case is:

$$\begin{aligned}
V_{it}^{ds}(\emptyset) = & -w_t f^s + x\beta \left[\bar{\pi}_{it+1}(\hat{\theta}_{ip'_k t}, 0) + (1-\delta)\mathbb{E}_{ip'_k t}^0 V_{it+1}(\hat{\theta}_{ip'_k t+1}, \tau_{ip'_k t+1}) + \delta V_{it+1}^\delta \right] \\
& + (1-x)\beta V_{it+1}^{ds}(\emptyset), \quad (2.18)
\end{aligned}$$

where the focal product line is vacant, and the continuation value after failed search is $V_{it+1}^{ds}(\emptyset)$. We assume that the firm's exit value per product line equals the value of a vacant line when both are evaluated simultaneously.¹⁵ In this case, it equals the value of waiting until $t + 1$ and conducting a search, as the firm exits only when all product lines are dropped and all searches fail, $V_{it}^\emptyset = \beta V_{it+1}^{ds}(\emptyset)$.¹⁶

Given the product-line value functions, we can characterize the search and product-drop cutoffs, $\mathcal{B}_{it}^{cs}(\tau_{ipt})$ and $\mathcal{B}_{it}^{ds}(\tau_{ipt})$. First, with posterior beliefs evaluated at the search cutoff, $\mathcal{B}_{it}(\hat{\theta}_{ip_k t}^{cs}, \tau_{ip_k t}) = \mathcal{B}_{it}^{cs}(\tau_{ip_k t})$, firms are indifferent between searching and not searching, which yields the following condition:

$$V_{it}^{ns}(\hat{\theta}_{ip_k t}^{cs}, \tau_{ip_k t}) = V_{it}^{cs}(\hat{\theta}_{ip_k t}^{cs}, \tau_{ip_k t}) \Rightarrow V_{it}^{ns}(\hat{\theta}_{ip_k t}^0, 0) - V_{it}^{ns}(\hat{\theta}_{ip_k t}^{cs}, \tau_{ip_k t}) = \frac{w_t f^s}{x}, \quad (2.19)$$

where $V_{it}^{ns}(\hat{\theta}_{ip_k t}^0, 0)$ is the term multiplied by x in $V_{it}^{cs}(\hat{\theta}_{ip_k t}^{cs}, \tau_{ip_k t})$. Thus, $(\hat{\theta}_{ip_k t}^{cs}, \tau_{ip_k t})$ is the belief that makes the difference between the continuation value of a newly drawn product under the initial posterior and the continuation value under the current posterior equal to the search cost scaled by the success probability.

In a similar fashion, evaluating posterior beliefs at the product-dropping cutoff, $(\hat{\theta}_{ip_k t}^{ds}, \tau_{ip_k t})$ satisfying $\mathcal{B}_{it}(\hat{\theta}_{ip_k t}^{ds}, \tau_{ip_k t}) = \mathcal{B}_{it}^{ds}(\tau_{ip_k t})$, firms are indifferent between search with product switching and product dropping. This gives the following condition:

$$V_{it}^{cs}(\hat{\theta}_{ip_k t}^{ds}, \tau_{ip_k t}) = V_{it}^{ds}(\emptyset). \quad (2.20)$$

If the firm fails to search in a vacant product line but does not exit, it initiates a new search in the following period. Since

$$V_{it}^{cs}(\hat{\theta}_{ip_k t}^{ds}, \tau_{ip_k t}) = -w_t f^s + x\beta \left[\bar{\pi}_{it+1}(\hat{\theta}_{ip_k t}^0, 0) + \mathbb{E}_{ip_k t}^0 V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) \right]$$

¹⁵Thus, in the case of exogenous exit, $V_{it+1}^\delta = V_{it+1}^{ds}(\emptyset)$, as exit occurs before product switching.

¹⁶This assumption is made for tractability. Under this assumption, the term following $(1-x)$ in $V_{it}^{ds}(\emptyset)$ simplifies to $(1 - \text{firm exit probability}) \times \beta V_{it+1}^{ds}(\emptyset) + (\text{firm exit probability}) \times \beta V_{it+1}^{ds}(\emptyset) = \beta V_{it+1}^{ds}(\emptyset)$, so the exit probability drops out of the value function.

$$+ (1 - x)\beta \left[\bar{\pi}_{it+1}(\hat{\theta}_{ip_k t}^{ds}, \tau_{ip_k t}) + \mathbb{E}_{ip_k t} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) \right],$$

it follows that

$$\bar{\pi}_{it+1}(\hat{\theta}_{ip_k t}^{ds}, \tau_{ip_k t}) + \mathbb{E}_{ip_k t} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) = V_{it+1}^{ds}(\emptyset).$$

To understand the functional properties of the two thresholds, we first analyze the monotonicity properties of product-line values. We begin by showing that higher posterior means $\hat{\theta}$ shift future beliefs strictly upward in the sense of first-order stochastic dominance.

Lemma 2 (First-Order Stochastic Dominance of Future Beliefs). *Fix (τ_t, a_t, φ_i) and suppose $\hat{\theta}_{t+1} \mid \hat{\theta}_t \sim N(\hat{\theta}_t, \hat{\sigma}_t^2 - \hat{\sigma}_{t+1}^2)$, where $\hat{\sigma}_t^2 - \hat{\sigma}_{t+1}^2 > 0$. Then, for any $\hat{\theta}_t^H > \hat{\theta}_t^L$, the conditional distribution $\hat{\theta}_{t+1} \mid \hat{\theta}_t^H$ first-order stochastically dominates $\hat{\theta}_{t+1} \mid \hat{\theta}_t^L$.*

Proof. Let $F_H(x) \equiv \Pr(\hat{\theta}_{t+1} \leq x \mid \hat{\theta}_t^H)$ and $F_L(x) \equiv \Pr(\hat{\theta}_{t+1} \leq x \mid \hat{\theta}_t^L)$ denote the corresponding conditional cumulative distribution functions. Since $\hat{\theta}_{t+1} \mid \hat{\theta}_t^j \sim N(\hat{\theta}_t^j, \hat{\sigma}_t^2 - \hat{\sigma}_{t+1}^2)$, $j \in \{H, L\}$, we have

$$F_j(x) = \Phi\left(\frac{x - \hat{\theta}_t^j}{\sqrt{\hat{\sigma}_t^2 - \hat{\sigma}_{t+1}^2}}\right),$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function.

Because $\hat{\theta}_t^H > \hat{\theta}_t^L$, it follows that for every x ,

$$\frac{x - \hat{\theta}_t^H}{\sqrt{\hat{\sigma}_t^2 - \hat{\sigma}_{t+1}^2}} < \frac{x - \hat{\theta}_t^L}{\sqrt{\hat{\sigma}_t^2 - \hat{\sigma}_{t+1}^2}}.$$

Since $\Phi(\cdot)$ is strictly increasing, $F_H(x) < F_L(x) \quad \forall x \in \mathbb{R}$. Hence,

$$\Pr(\hat{\theta}_{t+1} \leq x \mid \hat{\theta}_t^H) < \Pr(\hat{\theta}_{t+1} \leq x \mid \hat{\theta}_t^L) \quad \forall x \in \mathbb{R},$$

which implies that $\hat{\theta}_{t+1} \mid \hat{\theta}_t^H$ strictly first-order stochastically dominates $\hat{\theta}_{t+1} \mid \hat{\theta}_t^L$. $\square$

Because $\mathcal{B}(\hat{\theta}, \tau; a, \varphi)$ is strictly increasing in $\hat{\theta}$ and higher posterior means shift future

beliefs upward in the sense of strict first-order stochastic dominance, current profits are strictly increasing and expected continuation values are weakly increasing in $\hat{\theta}$. The following lemma establishes that V is weakly increasing in $\hat{\theta}$.

Lemma 3 (Monotonicity of Product-Line Values). *Fix (τ, a, φ) and prices. Suppose that $\mathcal{B}(\hat{\theta}, \tau; a, \varphi)$ is strictly increasing in $\hat{\theta}$. Then $V^{ns}(\hat{\theta}, \tau; a, \varphi)$, $V^{cs}(\hat{\theta}, \tau; a, \varphi)$ are strictly increasing in $\hat{\theta}$, while $V(\hat{\theta}, \tau; a, \varphi)$ are weakly increasing in $\hat{\theta}$, holding (τ, a, φ) fixed.*

Proof. The proof follows standard recursive methods arguments: the Bellman operator satisfies Blackwell's sufficient conditions and maps monotone functions into monotone functions. Hence, its unique fixed point inherits monotonicity; see [Stokey and Lucas Jr \(1989\)](#).

Let the state be $\xi = (\hat{\theta}, \tau, a, \varphi)$, and let $\mathcal{V}$ be the space of bounded continuous functions on the state space, equipped with the sup norm. Let $\mathcal{V}^+ \subset \mathcal{V}$ denote the set of functions that are weakly increasing in $\hat{\theta}$ for each fixed (τ, a, φ) . Define the no-search and search Bellman suboperators by

$$\begin{aligned} (T^{ns}V)(\xi) &= \beta\bar{\pi}(\xi) + \beta \int_{\mathcal{X}} V(\xi')g(\xi' | \xi)d\xi' \\ (T^{cs}V)(\xi) &= -wf^s + \beta(1-x) \left(\bar{\pi}(\xi) + \int_{\mathcal{X}} V(\xi')g(\xi' | \xi)d\xi' \right) \\ &\quad + \beta x \left(\bar{\pi}(\xi^0) + \int_{\mathcal{X}} V(\xi')g(\xi' | \xi^0)d\xi' \right), \end{aligned}$$

where ξ^0 denotes a state with $(\theta_0, 0)$. Then, the Bellman operator is

$$(TV)(\xi) = \max \{ (T^{ns}V)(\xi), (T^{cs}V)(\xi), V^{cs}(\emptyset; a, \psi, \varphi) \}.$$

Under bounded payoffs and $\beta \in (0, 1)$, it is easy to show that T satisfies Blackwell's sufficient conditions: it is monotone in V and discounts future values by β . Hence, by the contraction mapping theorem, T has a unique fixed point $V = T(V)$.

To establish the monotonicity of T , suppose $V, W \in \mathcal{V}$ satisfy $V(\xi) \leq W(\xi)$, $\forall \xi$. Then,

$$\int_{\mathcal{X}} V(\xi') g(\xi' | \xi) d\xi' \leq \int_{\mathcal{X}} W(\xi') g(\xi' | \xi) d\xi'.$$

Hence, $(T^{ns}V)(\xi) \leq (T^{ns}W)(\xi)$, and similarly, $(T^{ss}V)(\xi) \leq (T^{ss}W)(\xi)$. Taking the pointwise maximum preserves inequalities, so $(TV)(\xi) \leq (TW)(\xi)$. Thus, the Bellman operator is monotone.

For any constant $c \geq 0$, $(T^{ns}(V+c))(\xi) = (T^{ns}V)(\xi) + \beta c$, and similarly for T^{ss} . Therefore, $T(V+c) = TV + \beta c$. Since $\beta \in (0, 1)$, the Bellman operator satisfies Blackwell's discounting condition.

We now show that T maps $\mathcal{V}^+$ into itself. Take any $V \in \mathcal{V}^+$ and two states $\xi^H = (\hat{\theta}^H, \tau, a, \varphi)$, $\xi^L = (\hat{\theta}^L, \tau, a, \varphi)$, with $\hat{\theta}^H > \hat{\theta}^L$. Since $\mathcal{B}(\hat{\theta}, \tau; a, \varphi)$ is strictly increasing in $\hat{\theta}$, current operating profits are strictly higher at ξ^H than at ξ^L . For the continuation value, the non-belief components evolve according to the model's transition rules, while φ is fixed. Conditional on the same (τ, a, φ) , the only relevant difference between ξ^H and ξ^L is the distribution of next-period posterior beliefs. By Lemma 2, $\hat{\theta}' | \xi^H \succ_{FOSD} \hat{\theta}' | \xi^L$. Since $V \in \mathcal{V}^+$, $V(\hat{\theta}', \tau'; a', \varphi)$ is weakly increasing in $\hat{\theta}'$ for fixed (τ', a', φ) . Therefore, strict first-order stochastic dominance implies

$$\mathbb{E}[V(\hat{\theta}', \tau'; a', \varphi) | \xi^H] \geq \mathbb{E}[V(\hat{\theta}', \tau'; a', \varphi) | \xi^L].$$

Thus, expected continuation values are weakly increasing in $\hat{\theta}$. Since current operating profits are strictly increasing in $\hat{\theta}$, both $T^{ns}V$ and $T^{cs}V$ are strictly increasing in $\hat{\theta}$. The dropping value $V^{ss}(\emptyset; a, \varphi)$ is independent of $\hat{\theta}$ and is therefore weakly increasing in $\hat{\theta}$. Since the pointwise maximum of weakly increasing functions is weakly increasing, $TV \in \mathcal{V}^+$.

Therefore, T is a contraction and maps the closed set $\mathcal{V}^+$ into itself. The unique fixed point $V = T(V)$ must belong to $\mathcal{V}^+$. Hence $V(\hat{\theta}, \tau; a, \varphi)$ is weakly increasing in $\hat{\theta}$. Since V^{ns} and V^{cs} are the corresponding components of the Bellman operator evaluated at

the fixed point, they are strictly increasing in $\hat{\theta}$. $\square$

Lemma 4 (Dependence of Product-Line Values on Tenure). *Fix $(\hat{\theta}, a, \varphi)$ and prices. Suppose posterior variance $\hat{\sigma}^2(\tau, a, \varphi)$ varies with tenure τ . Then the product-line value $V(\hat{\theta}, \tau; a, \varphi)$ depends on τ .*

Proof. Since posterior variance is a function of τ , from (2.11), $\mathcal{B}(\hat{\theta}, \tau; a, \varphi)$ is a function of τ . Therefore, for fixed $(\hat{\theta}, a, \varphi)$, changes in τ alter $\mathcal{B}$. Since current operating profits are increasing in $\mathcal{B}$, current profits depend on τ .

Moreover, the transition kernel for future posterior beliefs depends on posterior variance, $\hat{\theta}' \mid (\hat{\theta}, \tau, a, \varphi) \sim N\left(\hat{\theta}, \hat{\sigma}^2(\tau, a, \varphi) - \hat{\sigma}'^2(\tau + 1, a + 1, \varphi)\right)$. Hence, tenure also affects the continuation value through future learning dynamics.

Therefore, both current payoffs and continuation values depend on τ , implying that $V(\hat{\theta}, \tau; a, \varphi)$ depends on tenure even conditional on $(\hat{\theta}, a, \varphi)$. $\square$

The following proposition establishes that the search and product-drop thresholds are uniquely determined by the product-line state $(\tau_{ipkt}, a_{it}, \varphi_i)$.

Proposition 1 (State Dependence of Search and Product-Drop Thresholds). *Suppose $\mathcal{B}(\hat{\theta}, \tau; a, \varphi)$ is strictly increasing in $\hat{\theta}$, and $V(\hat{\theta}, \tau; a, \varphi)$ is weakly increasing in $\hat{\theta}$ for fixed (τ, a, φ) . Then, for any given (τ, a, φ) , there exists a unique posterior mean $\hat{\theta}^{ds}(\tau, a, \varphi)$ satisfying the product-drop condition (2.20). Consequently, the product-drop threshold can be written as $\mathcal{B}_{ipkt}^{ds} \equiv \mathcal{B}^{ds}(\tau_{ipkt}, a_{it}, \varphi_i)$. Similarly, there exists a unique posterior mean $\hat{\theta}^{ss}(\tau, a, \varphi)$ satisfying the search condition (2.19), implying that the search threshold can be written as $\mathcal{B}_{ipkt}^{cs} \equiv \mathcal{B}^{cs}(\tau_{ipkt}, a_{it}, \varphi_i)$.*

Proof. By Lemma 3, $V(\hat{\theta}, \tau; a, \varphi)$ is weakly increasing in $\hat{\theta}$ for fixed (τ, a, φ) . Since $\mathcal{B}(\hat{\theta}, \tau; a, \varphi)$ is strictly increasing in $\hat{\theta}$, the corresponding continuation values inherit monotonicity in $\hat{\theta}$. Therefore, the left-hand side of the product-drop condition (2.20) is monotone in $\hat{\theta}$ holding (τ, a, φ) fixed, while the right-hand side is independent of $\hat{\theta}$. Hence, there exists a unique posterior mean $\hat{\theta}^{ds}(\tau, a, \varphi)$ satisfying the indifference

condition. Because $\mathcal{B}(\hat{\theta}, \tau; a, \varphi)$ is one-to-one in $\hat{\theta}$ conditional on (τ, a, φ) , this uniquely defines a product-drop threshold $\mathcal{B}^{ds}(\tau, a, \varphi)$.

An identical argument applies to the search condition (2.19), which implies the existence of a unique posterior mean $\hat{\theta}^{cs}(\tau, a, \varphi)$ satisfying the search indifference condition, which uniquely determines the search threshold $\mathcal{B}^{cs}(\tau, a, \varphi)$. $\square$

Lastly, we characterize conditions under which $\mathcal{B}_{it}^{cs}(\tau) > \mathcal{B}_{it}^{ds}(\tau)$. Since $V_{it}^{ns}(\hat{\theta}, \tau)$ and $V_{it}^{cs}(\hat{\theta}, \tau)$ are strictly increasing in $\hat{\theta}$, it follows that $V_{it}^{ns}(\hat{\theta}, \tau) > V_{it}^{cs}(\hat{\theta}, \tau)$ for all $\hat{\theta} > \hat{\theta}^{cs}$, and $V_{it}^{cs}(\hat{\theta}, \tau) > V_{it}^{ds}(\emptyset)$ for all $\hat{\theta} > \hat{\theta}^{ds}$. Therefore, a sufficient condition for $\mathcal{B}_{it}^{cs}(\tau) > \mathcal{B}_{it}^{ds}(\tau)$ is $V_{it}^{ns}(\hat{\theta}^{ds}, \tau) < V_{it}^{cs}(\hat{\theta}^{ds}, \tau)$, that is, firms strictly prefer searching while retaining the current product to continuing production without search at the drop threshold. Using the Bellman equations, this condition is equivalent to

$$\underbrace{\bar{\pi}_{it}(\hat{\theta}_{ipt}^0, 0) + \mathbb{E}_{ipt}^0 V_{it+1}^0(\hat{\theta}', \tau')}_{\text{new product value}} - \underbrace{(\bar{\pi}_{it}(\hat{\theta}_{ipt}^{ds}, \tau_{ipt}) + \mathbb{E}_{ipt} V_{it+1}(\hat{\theta}', \tau'))}_{\text{continuation value of marginal existing product}} > \frac{w_t f^s}{\beta x}. \quad (2.21)$$

This condition states that the expected gain from successfully replacing the current product must be sufficiently large relative to the search cost. Hence, $\mathcal{B}_{it}^{cs}(\tau) > \mathcal{B}_{it}^{ds}(\tau)$ is more likely to hold when search is more effective (high x), search costs are lower, or newly drawn products have substantially higher continuation values than marginal existing products. This generates an intermediate region in which firms continue producing existing products while simultaneously searching for replacement products.

2.3.6 Firm Value Function

Let $\Theta_{it} \equiv \left\{ \left(\hat{\theta}_{ip_k t}, \tau_{ip_k t} \right) \right\}_{k=1}^{n_i}$ denote firm i 's portfolio of posterior beliefs about product-specific expertise, where k indexes product lines and p_k denotes product p assigned to product line k . The firm's exit probability in the next period equals the probability that all posterior beliefs $\left\{ \mathcal{B}_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) \right\}_{k=1}^{n_i}$ fall below the exit cutoff and all searches

fail:

$$(1 - x)^{n_i} \prod_{k=1}^{n_i} \Pr \left(\mathcal{B}_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) < \mathcal{B}_{it+1}^d(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) \mid \mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) \right).$$

For vacant lines, this conditional probability degenerates to one, as a failed search leaves the line vacant.

Under Bayesian learning, these conditional probabilities do not reduce to unconditional probabilities. Consequently, product-line values are interdependent: posterior beliefs about product-specific expertise in other lines affect the exit probability and, hence, the value of product line k . We remove this dependence by assuming that the value of exit equals the value of dropping the product and continuing to search.¹⁷ Under this assumption, the following lemma establishes that product-line values are independent across lines:

Lemma 5 (Independence of Product Line Values). *Suppose that the value of a product line upon firm exit equals the value of dropping the product while continuing to search, i.e., $V^\emptyset(a_{it}, \psi_{it}, \varphi_i) = \beta V^{ss}(\emptyset; a_{it+1}, \psi_{it+1}, \varphi_i)$. Then the value of product line k depends on its own state, $(\hat{\theta}_{ip_k t}, \tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i)$, and is independent of match qualities in other product lines, $V(\hat{\theta}_{ip_k t}, \tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i)$.*

Proof. The value of product line k is determined by flow profits, the switching decision, and the firm's exit decision. Flow profits depend only on $(\hat{\theta}_{ip_k t}, \hat{\sigma}_{ip_k t}, \theta_{ip_k t}, \varphi_i)$, and the evolution of $(\hat{\theta}_{ip_k t}, \hat{\sigma}_{ip_k t}, \theta_{ip_k t})$ depends only on firm-level states $(a_{it}, \psi_{it}, \varphi_i)$ and product-specific factors.

When $\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) < \mathcal{B}_{it}^d(\hat{\theta}_{ipt}, \tau_{ipt})$, the continuation value depends on whether the firm exits or continues to search. By assumption, both outcomes yield the same continuation value $\beta V^{ss}(\emptyset; a_{it+1}, \psi_{it+1}, \varphi_i)$. Therefore, although the probability of exit depends on the evolution of match qualities in other product lines, this dependence does not affect the value of product line k , since exit and continuation are payoff-equivalent.

¹⁷This assumption also ensures that firms optimally exit when all products they produce fall below the drop threshold.

It follows that the value of product line k depends only on $(\hat{\theta}_{ip_k t}, \tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i)$ and not on Θ_{it} . This establishes independence across product lines. $\square$

As a result, the firm value is the sum of the values of its product lines:

Proposition 2. *The firm-level value function is additively separable across product lines. In particular,*

$$V(\Theta_{it}; a_{it}, \psi_{it}, \varphi_i) = \sum_{k=1}^{n_i} V(\hat{\theta}_{ip_k t}, \tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i).$$

Proof. Follows from Lemma 5. $\square$

2.4 Potential Startups

Each period, a large pool of potential startups makes entry decisions concurrently with incumbents' search decisions. Potential startups draw firm type φ_i from a stationary distribution $H(\varphi)$. Prior to entry, firms do not observe their realization of φ_i and make entry decisions based on the expected value of entry. Entry is undirected, and each potential startup enters only one product market, with the assigned market realized at the end of the period. Entry incurs a fixed cost f^e , measured in units of labor. Upon entry, φ_i is revealed. After entry and initial production, age-one startups conduct multiple searches up to their capacity $n(\varphi)$. Since they initially operate in only one product market, they search over a set of size $N - 1$, with assignments coordinated without replacement as before.¹⁸

Then, free entry implies that the expected value of entry equals the entry cost for all startups:

$$\mathbb{E}_{\varphi} [V_t^e(\varphi)] = w_t f^e, \quad (2.22)$$

where

$$V_t^e(\varphi) = \max \left(\beta \left[\bar{\pi}(\hat{\theta}_{ip_1 t}^0, 0; 0, \psi_{it}, \varphi) + \mathbb{E}_{(\hat{\theta}_{ip_1 t+1}, \tau_{ip_1 t+1}) | (\hat{\theta}_{ip_1 t}^0, 0)} V_{t+1}(\hat{\theta}_{ip_1 t+1}, \tau_{ip_1 t+1}; 1, \psi_{it+1}, \varphi) \right] \right)$$

¹⁸Under uniform product draws and symmetric product markets, expectations over product assignments collapse. We therefore suppress them in the value function.

$$+ \beta (n(\varphi) - 1) V_{t+1}^{ds}(\emptyset; 1, \psi_{it+1}, \varphi), 0 \Big). \Big)$$

The first line represents the value of the startup's initial product line, which begins with the initial prior belief over product-specific expertise. The first term in the second line represents the value of $n(\varphi) - 1$ additional searches associated with the startup's initially vacant product lines. Furthermore, upon entry, firms with low realizations of φ_i exit immediately if their continuation value is negative. This is reflected on the max operator with zero value in the second line. Thus, there exists an exit cutoff $\underline{\varphi}^e$ such that

$$V_t^e(\underline{\varphi}^e) = 0. \quad (2.23)$$

2.5 Equilibrium

2.5.1 Labor Market Clearing

From (2.10), the optimal labor demand of firm i for product p at $t + 1$ is

$$\ell_{ipt+1}(\hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i) = \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt+1}^\eta C_{pt+1} \left[\mathcal{B}(\hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i) \right]^\eta z(\varphi_i)^{\eta-1} w_{t+1}^{-\eta}. \quad (2.24)$$

Thus, total labor demand for product p is

$$L_{pt+1} = \mathcal{M}_{pt} \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt+1}^\eta C_{pt+1} w_{t+1}^{-\eta} \int_{\mathcal{X}} \mathcal{B}(\xi)^\eta z(\varphi)^{\eta-1} \bar{\mu}_{pt}(\xi) d\xi, \quad (2.25)$$

where $\mathcal{M}_{pt}$ denotes the mass of firms (equivalently, product lines) operating in product market p at the end of period t (equivalently, at the beginning of $t + 1$), $\bar{\mu}_{pt}$ denotes the corresponding product-line density over states $\xi \equiv (\hat{\theta}, \tau, a, \varphi)$, and $\mathcal{X} \equiv \mathbb{R} \times \mathbb{N} \times \mathbb{N} \times \Phi$ is the product-line state space, excluding the vacant lines, where $\Phi \subset \mathbb{R}$ is a (compact) set indexing firm types. The labor market clearing condition is

$$\bar{L} = \sum_{p=1}^N L_{pt+1} + \mathcal{F}_{t+1}^o + \mathcal{F}_{t+1}^s + \mathcal{F}_{t+1}^e, \quad (2.26)$$

where $\mathcal{F}_{t+1}^o$, $\mathcal{F}_{t+1}^s$, and $\mathcal{F}_{t+1}^e$ denote total labor used for operating, switching, and entry fixed costs, respectively, as will be defined in Section 2.5.3.

2.5.2 Product-Line Distribution and Firm Distribution

Let $\mu_{pt}(A) \equiv \mathcal{M}_{pt}\bar{\mu}_{pt}(A)$ denote the product-line distribution measuring total mass of firms (equivalently, product lines) in market p with states lying in $A \subseteq \mathcal{X}$, measured at the end of period t (equivalently, at the beginning of $t + 1$) after the realization of θ_{ipt} and search outcomes. Then, for $\tau' > 0$, the law of motion at state $(\hat{\theta}', \tau', a', \varphi)$, corresponding to incumbent product lines that remain in product market p , is given by

$$\begin{aligned} \mu_{p,t+1}(\hat{\theta}', \tau', a', \varphi) &= (1 - \delta) \int_{\mathbb{R}} \left[\mathbf{1}_{\mathcal{R}_{t+1}^{ns}}(\hat{\theta}', \tau', a', \varphi) + (1 - x) \mathbf{1}_{\mathcal{R}_{t+1}^{cs}}(\hat{\theta}', \tau', a', \varphi) \right] \\ &\quad \times g(\hat{\theta}', \tau', a', \varphi \mid \hat{\theta}, \tau' - 1, a' - 1, \varphi) \mu_{pt}(\hat{\theta}, \tau' - 1, a' - 1, \varphi) d\hat{\theta}. \end{aligned} \quad (2.27)$$

For $\tau' = 0$, $(\hat{\theta}_{ipt+1}, \tau_{ipt+1}, a_{ipt+1}, \varphi_i) = (\hat{\theta}^0, 0, a', \varphi)$ when incumbent firm i successfully switches products at the end of $t + 1$ and is assigned to product market p , including through previously vacant product lines, or when potential startups enter market p :

$$\begin{aligned} \mu_{p,t+1}(\hat{\theta}^0, 0, a', \varphi) &= (1 - \delta) \sum_{q \neq p} \sum_{\check{\tau}'} \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{x}{N - 1} \mathbf{1}_{\mathcal{R}_{t+1}^{cs} \cup \mathcal{R}_{t+1}^{ds}}(\hat{\theta}', \check{\tau}', a', \varphi) \\ &\quad \times g(\hat{\theta}', \check{\tau}', a', \varphi \mid \hat{\theta}, \check{\tau}' - 1, a' - 1, \varphi) \mu_{qt}(\hat{\theta}, \check{\tau}' - 1, a' - 1, \varphi) d\hat{\theta} d\check{\tau}' \\ &\quad + (1 - \delta) \sum_{q \neq p} \frac{x}{N - 1} \mu_{qt}(\emptyset, a' - 1, \varphi) \\ &\quad + \frac{\mathcal{M}_{t+1}^e}{N} h(\varphi) \mathbf{1}\{a' = 0\}. \end{aligned} \quad (2.28)$$

$\frac{1}{N-1}$ is the adjusted probability of being assigned to product market p .¹⁹ h denotes the

¹⁹For firms with $a' \geq 2$ and φ , the feasible search set has size $N - n(\varphi)$. Hence a product line currently assigned to q lands in p with probability $\frac{1}{N - n(\varphi)}$. However, if the firm already operates in p through another line, this probability is zero. For a firm with scope $n(\varphi)$ that operates in q , the probability that it also operates in $p \neq q$ is $\frac{n(\varphi)-1}{N-1}$ under random assignment without replacement. Taking this into

density associated with H , and $\mu_{qt}(\emptyset, a' - 1, \varphi)$ denotes the mass of vacant product lines associated with product q belonging to firms of age $a' - 1$ and type φ . Since entry is undirected, each product market receives $\mathcal{M}_{t+1}^e/N$ mass of startups.

Lastly, the mass of vacant product lines associated with product market p evolves according to

$$\begin{aligned} \mu_{pt+1}(\emptyset, a', \varphi) &= (1 - \delta)(1 - p^e(a', \varphi)) \sum_{q \neq p} \sum_{\check{\tau}'} \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{1 - x}{N - 1} \mathbf{1}_{\mathcal{R}_{t+1}^{ds}}(\hat{\theta}', \check{\tau}', a', \varphi) \\ &\quad \times g(\hat{\theta}', \check{\tau}', a', \varphi \mid \hat{\theta}, \check{\tau}' - 1, a' - 1, \varphi) \mu_{qt}(\hat{\theta}, \check{\tau}' - 1, a' - 1, \varphi) d\hat{\theta} d\hat{\theta}' \\ &\quad + (1 - \delta)(1 - p^e(a', \varphi)) \sum_{q \neq p} \frac{1 - x}{N - 1} \mu_{qt}(\emptyset, a' - 1, \varphi), \end{aligned} \quad (2.29)$$

where $p^e(a', \varphi) \equiv [q^\emptyset(a', \varphi)]^{n(\varphi)-1}$ denotes the approximate probability that a firm of age a' and type φ has all non-focal product lines vacant and that all associated searches fail.²⁰ $q^\emptyset(a', \varphi)$ denote the generic probability that a product line not associated with product p is vacant and the corresponding search fails at the end of period $t + 1$:

$$\begin{aligned} q^\emptyset(a', \varphi) &= \sum_{q \neq p} \frac{1}{N - 1} \left[\bar{\mu}_{qt}(\emptyset, a' - 1, \varphi) + \sum_{\check{q} \neq q} \sum_{\check{\tau}'} \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{1}{N - 1} \mathbf{1}_{\mathcal{R}_{t+1}^{ds}}(\hat{\theta}', \check{\tau}', a', \varphi) \right. \\ &\quad \left. \times g(\hat{\theta}', \check{\tau}', a', \varphi \mid \hat{\theta}, \check{\tau}' - 1, a' - 1, \varphi) \bar{\mu}_{\check{q}t}(\hat{\theta}, \check{\tau}' - 1, a' - 1, \varphi) d\hat{\theta} d\hat{\theta}' \right] (1 - x). \end{aligned}$$

Thus, $1 - p^e(a', \varphi)$ is the probability that the firm does not exit endogenously, so that the vacant product line remains in the state space. The two terms in (2.29) respectively capture transitions into vacancy from initially occupied and initially vacant product lines associated with markets other than p .

Let the firm-level state be $\xi_{it}^f = (\Theta_{it}, a_{it}, \varphi_i)$, where $\Theta_{it} \equiv \{(\hat{\theta}_{ip_k t}, \tau_{ip_k t})\}_{k=1}^{n(\varphi_i)}$. Let

account, the adjusted probability that a product line in q lands in p is $\frac{1}{N - n(\varphi)} \times \frac{N - n(\varphi)}{N - 1} = \frac{1}{N - 1}$ by a law of large numbers with a continuum of firms. For firms with $a' = 1$ operating in q , the feasible search set has size $N - 1$, and the firm cannot already operate in p . Hence, the assignment probability is also $\frac{1}{N - 1}$.

²⁰While this event depends on the joint distribution of product-line states within a firm, tracking it explicitly would require a high-dimensional firm-level state space. To maintain tractability, we approximate the joint probability using the cross-sectional product-line density $\bar{\mu}_{pt}$. We assess the accuracy of this approximation in the simulation stage.

$\Gamma_t(\xi^f)$ denote the firm-level cross-sectional distribution. The product-line distribution μ_{pt} is obtained by projecting the firm-level distribution Γ_t onto the space of product lines grouped by product markets. Thus the product-line distribution and the firm distribution are linked through a projection map, and scope heterogeneity enters only through the number of product-line distributions to which each firm contributes. Because the firm-level distribution is a high-dimensional object and does not directly enter the equilibrium conditions, we track the product-line distribution and recover the firm distribution in the simulation stage. The total mass of product lines (equivalently, firms) in market p is obtained by integrating the distributions in (2.27)-(2.29) over all states.

2.5.3 Aggregate Variables

Since there is $\mathcal{M}_{pt-1}$ mass of firms producing product p in period t , total labor used for operating fixed costs is

$$\mathcal{F}_t^o = f^o \sum_{p=1}^N \mathcal{M}_{pt-1}. \quad (2.30)$$

Total labor used for search fixed costs in period t is

$$\begin{aligned} \mathcal{F}_t^s = f^s \sum_{p=1}^N \mathcal{M}_{pt-1} & \left[\int_{\mathcal{X}} \int_{\mathcal{X}} \mathbf{1}_{\mathcal{R}_t^{cs} \cup \mathcal{R}_t^{ds}}(\xi_t) g(\xi_t \mid \xi_{t-1}) \bar{\mu}_{pt-1}(\xi_{t-1}) d\xi_{t-1} d\xi_t \right. \\ & \left. + \int_{(a, \varphi)} \bar{\mu}_{pt-1}(\emptyset, a, \varphi) d(a, \varphi) \right], \end{aligned} \quad (2.31)$$

and total labor used for entry fixed costs in period t is

$$\mathcal{F}_t^e = f^e \mathcal{M}_t^e (1 + H(\underline{\varphi}^e)), \quad (2.32)$$

where $H(\underline{\varphi}^e)$ is the share of startups that immediately exit because their productivity falls below the threshold value.

To characterize the product-level price index P_{pt} , define the generic counterparts of $(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \varphi_i)$, θ_{ipt} , and θ_{ip} as $(\hat{\theta}_{pt}, \tau, a, \varphi)$, θ_{pt} , and θ_p , respectively. Since the

market-clearing price for a product line in state ξ_{pt-1} with realization θ_{pt} is $p_{pt}(\xi_{pt-1}) = \frac{\eta}{\eta-1} \frac{w_t}{z(\varphi)} \frac{\theta_{pt}^{1/\eta}}{\mathcal{B}(\xi_{pt-1})}$, computing the price index requires the conditional distribution of θ_{pt} given ξ_{pt-1} . Since the initial prior distribution of θ_p coincides with the true data-generating process and firms use a correctly specified Bayesian model, the posterior distribution coincides with the objective conditional distribution. Hence, conditional on state ξ_{pt-1} , $\theta_{pt} \mid \xi_{pt-1} \sim \text{LogNormal}(\hat{\theta}_{pt-1}, \hat{\sigma}_{pt-1}^2 + \sigma_\varepsilon^2)$. Because there is a continuum of product lines in each state, idiosyncratic realizations of θ_{pt} average out in the aggregate. Hence, the aggregate value of $\theta_{pt}^{1/\eta}$ is equal to its conditional expectation, $\mathbb{E}[\theta_{pt}^{1/\eta} \mid \xi_{pt-1}] = \mathcal{B}(\xi_{pt-1})$. The product-level price index can therefore be written as²¹

$$P_{pt} = \frac{\eta}{\eta-1} w_t \left(\mathcal{M}_{pt-1} \int_{\mathcal{X}} \mathcal{B}(\xi_{pt-1})^\eta z(\varphi)^{\eta-1} \bar{\mu}_{pt-1}(\xi_{pt-1}) d\xi_{pt-1} \right)^{-\frac{1}{\eta-1}}. \quad (2.33)$$

Using (2.25) and (2.26), we can derive the value of C_{pt} in a symmetric equilibrium. In a symmetric equilibrium, total labor demand is identical across product markets. Thus, from (2.26), $L_{pt} = \frac{1}{N} (\bar{L} - \mathcal{F}_t^o - \mathcal{F}_t^s - \mathcal{F}_t^e)$. Substituting this expression into (2.25), plugging in (2.33), and rearranging yields

$$C_{pt} = \frac{\bar{L} - \mathcal{F}_t^o - \mathcal{F}_t^s - \mathcal{F}_t^e}{N} \left(\mathcal{M}_{pt-1} \int_{\mathcal{X}} \mathcal{B}(\xi_{pt-1})^\eta z(\varphi)^{\eta-1} \bar{\mu}_{pt-1}(\xi_{pt-1}) d\xi_{pt-1} \right)^{\frac{1}{\eta-1}}. \quad (2.34)$$

Then, using (2.10), (2.33), and (2.34), real GDP, $Y_t \equiv \sum_{p=1}^N \int_{\mathcal{X}} y_{pt}^*(\xi) \mu_{pt-1}^*(\xi) d\xi$ can

²¹Let $f(\theta_{pt} \mid \xi_{pt-1})$ denote the conditional density of θ_{pt} . Then,

$$\begin{aligned} P_{pt} &= \left(\int_{\mathcal{X}} \int_{\mathbb{R}} \theta_{pt} p_{pt}(\xi_{pt-1})^{1-\eta} f(\theta_{pt} \mid \xi_{pt-1}) \mu_{pt-1}(\xi_{pt-1}) d\theta_{pt} d\xi_{pt-1} \right)^{\frac{1}{1-\eta}} \\ &= \frac{\eta}{\eta-1} w_t \left[\int_{\mathcal{X}} \mathcal{B}(\xi_{pt-1})^{\eta-1} z(\varphi)^{\eta-1} \left(\int_{\mathbb{R}} \theta_{pt}^{1/\eta} f(\theta_{pt} \mid \xi_{pt-1}) d\theta_{pt} \right) \mu_{pt-1}(\xi_{pt-1}) d\xi_{pt-1} \right]^{\frac{1}{1-\eta}} \\ &= \frac{\eta}{\eta-1} w_t \left(\int_{\mathcal{X}} \mathcal{B}(\xi_{pt-1})^\eta z(\varphi)^{\eta-1} \mu_{pt-1}(\xi_{pt-1}) d\xi_{pt-1} \right)^{\frac{1}{1-\eta}}. \end{aligned}$$

be written as

$$Y_t = \frac{\bar{L} - \mathcal{F}_t^o - \mathcal{F}_t^s - \mathcal{F}_t^e}{N} \sum_{p=1}^N \frac{\int_{\mathcal{X}} \mathcal{B}(\xi)^\eta z(\varphi)^\eta \bar{\mu}_{pt-1}(\xi) d\xi}{\int_{\mathcal{X}} \mathcal{B}(\xi)^\eta z(\varphi)^{\eta-1} \bar{\mu}_{pt-1}(\xi) d\xi}.$$

2.5.4 Symmetric Stationary Recursive Competitive Equilibrium

Definition 1. A symmetric stationary recursive competitive equilibrium of this economy is a tuple

$$\{C, P, (\hat{\theta}, \hat{\sigma}^2), y, p, \ell, V, V^{ns}, V^{cs}, V^{ds}, \mathcal{B}^{cs}, \mathcal{B}^{ds}, V^e, \underline{\varphi}^e, L, \mathcal{F}^o, \mathcal{F}^s, \mathcal{F}^e, w, \mu\}$$

such that: (i) C and P satisfy (2.34) and (2.33) in all product markets; (ii) posterior beliefs $(\hat{\theta}, \hat{\sigma}^2)$ about product-specific expertise satisfy (2.6) and (2.7); (iii) given C , P , w and $(\hat{\theta}, \hat{\sigma}^2)$, output y satisfying (2.10) maximizes profits in (2.9), the market-clearing price p satisfies (2.12), and labor demand ℓ satisfies (2.24) in all product markets; (iv) the value functions V , V^{ns} , V^{cs} , and V^{ds} satisfy (2.15), (2.16), (2.17) and (2.18); (v) the search and drop thresholds $\mathcal{B}^{cs}$ and $\mathcal{B}^{ds}$ satisfy (2.19) and (2.20); (vi) V^e satisfies the free-entry condition (2.22), and $\underline{\varphi}^e$ satisfies (2.23); (vii) L , $\mathcal{F}^o$, $\mathcal{F}^{ss}$, and $\mathcal{F}^e$ defined by (2.25), (2.30), (2.31), and (2.32), together with wage w , jointly satisfy the labor market clearing condition (2.26); (viii) the distribution μ is invariant under the law of motion induced by firms' policies, Bayesian updating, entry, and exit as characterized by (2.27)-(2.29); and (ix) all product markets are identical.

2.6 Model Implications

2.6.1 Learning Frictions and Production

Building on the biased learning result in Lemma 1, we define a wedge between perceived and true product-specific expertise to characterize how beliefs affect production decisions. This wedge captures how firms' posterior beliefs enter output through $\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1})$, allowing firm-level output to be expressed as a scaled version of the

level implied by θ_{ip} . Because firms with lower information-processing ability—when they are young—place greater weight on the prior and face noisier signals, they systematically underestimate high- θ_{ip} products and overestimate low- θ_{ip} products. As a result, these firms produce relatively less of high- θ_{ip} products and relatively more of low- θ_{ip} products compared to firms with higher ψ . The following proposition formalizes this relationship.

Proposition 3 (Learning Frictions and Production). *Define the wedge between perceived and true product-specific fundamentals as*

$$1 - \omega_{ipt} \equiv \frac{\left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta}{\theta_{ip}}. \quad (2.35)$$

Then firm-level output can be written as

$$y_{ipt}^* = (1 - \omega_{ipt}) \theta_{ip} z(\varphi_i)^\eta \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt}^\eta C_{pt} w_t^{-\eta}.$$

Firms with lower information-processing ability ψ_{it} exhibit systematic variation in ω_{ipt} : for a wider range of high- θ_{ip} products, $\omega_{ipt} > 0$, while for low- θ_{ip} products, $\omega_{ipt} < 0$. As a result, these firms produce relatively less of high- θ_{ip} products and relatively more of low- θ_{ip} products compared to firms with higher ψ_{it} .

Proof. Recall that the optimal output for product p by firm i with productivity φ_i , given beliefs $(\hat{\theta}_{ipt-1}, \tau_{ipt-1})$ in period t , is

$$y_{ipt}^* = \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt}^\eta C_{pt} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta z(\varphi_i)^\eta w_t^{-\eta},$$

where $\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) = \exp \left\{ \frac{\hat{\theta}_{ipt-1}}{\eta} + \frac{1}{2} \left(\frac{\hat{\sigma}_{ipt-1}^2 + \sigma_\varepsilon^2}{\eta^2} \right) \right\}$, the posterior variance is given by $\hat{\sigma}_{ipt-1}^2 = \left(\frac{1}{\sigma_0^2} + \frac{\tau_{ipt-1}}{\sigma_\varepsilon^2} + \frac{\sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} \right)^{-1}$, and the posterior mean can be written as

$$\hat{\theta}_{ipt-1} = \hat{\sigma}_{ipt-1}^2 \left[\frac{\theta_0}{\sigma_0^2} + \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} \log \theta_{ip} + \frac{1}{\sigma_\varepsilon^2} \left(\sum_{\tilde{t}} \varepsilon_{ip\tilde{t}} + \sum_{\tilde{t}} \psi_{i\tilde{t}} \nu_{ip\tilde{t}} \right) \right].$$

Taking expectations, we obtain the average value of $\left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1})\right]^\eta$ as

$$\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta = \exp \left[\hat{\sigma}_{ipt-1}^2 \left(\frac{\theta_0}{\sigma_0^2} + \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} \log \theta_{ip} \right) + \frac{1}{2} (\hat{\sigma}_{ipt-1}^2)^2 \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} + \frac{1}{2} \frac{\hat{\sigma}_{ipt-1}^2 + \sigma_\varepsilon^2}{\eta} \right].$$

Let $A \equiv \frac{1}{\sigma_0^2}$ and $B \equiv \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2}$. Then

$$\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta = \exp \left[\frac{A\theta_0 + B \log \theta_{ip}}{A + B} + \frac{B}{2(A + B)^2} + \frac{1}{2\eta} \left(\frac{1}{A + B} + \sigma_\varepsilon^2 \right) \right].$$

As $\psi_{it} \rightarrow \infty$ for all t , signals become perfectly informative and

$$\left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta \rightarrow \theta_{ip},$$

even with a finite sample.

To characterize wedges, define the threshold

$$\theta_{ip}^* = \exp \left[\theta_0 + \frac{1}{2A} \left(\frac{B}{A + B} + \frac{1}{\eta} + \frac{A + B}{\eta} \sigma_\varepsilon^2 \right) \right].$$

Then, $\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta > \theta_{ip}$ if $\theta_{ip} < \theta_{ip}^*$, and $\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta < \theta_{ip}$ if $\theta_{ip} > \theta_{ip}^*$. Moreover, since $\frac{\partial \log \theta_{ip}^*}{\partial B} > 0$, firms with lower ψ_{it} have a lower threshold. Thus, low- ψ firms underestimate $\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta$ for a wider range of high θ_{ip} .

Define the wedge

$$1 - \omega_{ipt} \equiv \frac{[\mathcal{B}_{it-1}]^\eta}{\theta_{ip}}.$$

Then output can be written as

$$y_{ipt}^* = (1 - \omega_{ipt}) \theta_{ip} z(\varphi_i)^\eta \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt}^\eta C_{pt} w_t^{-\eta}.$$

This implies that low- ψ firms underproduce high- θ_{ip} products and overproduce low- θ_{ip}

products relative to the allocation implied by the true θ_{ip} . □

2.6.2 Learning Frictions and Product Switching

In addition to its effect on output, learning also influences firms' product-switching decisions by affecting the two switching thresholds. To illustrate this mechanism more transparently, and for expositional simplicity, we consider a two-period environment in which firms operate a single product and enter period $t=1$ with given posterior beliefs.

In period 1, after the realization of θ_{ip1} , firms update their posterior beliefs to $(\hat{\theta}_{ip1}, 1)$. They then decide whether to search, and conditional on the search outcome, choose quantities for the product operated in the next period. Period-2 profits are determined by the static profit maximization problem and therefore coincide with (2.14). Using this setup, we characterize the firm's value function in period at the search decision stage (after the realization of θ_{ip1}) as follows:

$$V_{i1}(\hat{\theta}_{ip1}, 1) = \max \left\{ V_{i1}^{ns}(\hat{\theta}_{ip1}, 1), V_{i1}^{cs}(\hat{\theta}_{ip1}, 1), V_{i1}^{ds}(\emptyset) \right\},$$

where

$$\begin{aligned} V_{i1}^{ns}(\hat{\theta}_{ip1}, 1) &= \beta \mathbb{E}_{\theta_{ip2}} \pi_{i2}(\hat{\theta}_{ip1}, 1) \\ V_{i1}^{cs}(\hat{\theta}_{ip1}, 1) &= -w_1 f^s + \beta \left(x \mathbb{E}_{\theta_{ip2}} \pi_{i2}(\hat{\theta}^0, 0) + (1-x) \mathbb{E}_{\theta_{ip2}} \pi_{i2}(\hat{\theta}_{ip1}, 1) \right) \\ V_{i1}^{ds}(\emptyset) &= -w_1 f^s + \beta x \mathbb{E}_{\theta_{ip2}} \pi_{i2}(\hat{\theta}^0, 0). \end{aligned}$$

Under the assumption that $\mathcal{B}_{i1}(\hat{\theta}^0, 0) > \mathcal{B}_{i1}^{ds}$, we obtain the following proposition, which characterizes product-switching decisions as a function of firms' learning ability ψ_{i1} and productivity $z(\varphi_i)$.

Proposition 4 (Learning Frictions and Product Switching). *Firms with lower information-processing ability ψ_i are more prone to operating in misaligned product lines due to learning. When the true θ_{ip} is high, they are more likely to replace or drop products that should*

be retained; whereas when θ_{ip} is low, they are more likely to retain products that should be dropped. Furthermore, this misalignment becomes more pronounced among firms with higher productivity $z(\varphi_i)$, i.e., the likelihood that firms operate misaligned product lines for a given ψ_i increases in $z(\varphi_i)$.

Proof. The firm's problem can be characterized by three cases:

$$V_{i1}(\hat{\theta}_{ip1}, 1) = \begin{cases} -w_1 f^s + \beta \left[x(\Xi_{i2} \mathcal{B}_{i1}(\hat{\theta}^0, 0)^\eta - w_2 f^o) \right. \\ \quad \left. + (1-x)(\Xi_{i2} \mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)^\eta - w_2 f^o) \right] & , \text{ if } \mathcal{B}_{i1}^{cs} > \mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) \geq \mathcal{B}_{i1}^{ds} \\ -w_1 f^s + \beta x \left[\Xi_{i2} \mathcal{B}_{i1}(\hat{\theta}^0, 0)^\eta - w_2 f^o \right] & , \text{ if } \mathcal{B}_{i1}^{ds} > \mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) \\ \beta \left[\Xi_{i2} \mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)^\eta - w_2 f^o \right] & , \text{ if } \mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) \geq \mathcal{B}_{i1}^{cs}, \end{cases}$$

where $\Xi_{i2} \equiv \frac{(\eta-1)^{\eta-1}}{\eta^\eta} \left(\frac{z(\varphi_i)}{w_2} \right)^{\eta-1} P_{p2}^\eta C_{p2}$. The first two cases correspond to search, while the last corresponds to no search.

When $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) = \mathcal{B}_{i1}^{cs}$, the firm is indifferent between searching and not searching. Thus,

$$\mathcal{B}_{i1}^{cs} = \left(\mathcal{B}(\hat{\theta}^0, 0)^\eta - \frac{w_1 f^s}{\beta x \Xi_{i2}} \right)^{\frac{1}{\eta}}. \quad (2.36)$$

Since Ξ_{i2} is increasing in $z(\varphi_i)$, the search threshold $\mathcal{B}_{i1}^{cs}$ is increasing in $z(\varphi_i)$.

When $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) = \mathcal{B}_{i1}^{ds}$, the firm is indifferent between retaining and dropping the product. Thus,

$$\mathcal{B}_{i1}^{ds} = \left(\frac{w_2 f^o}{\Xi_{i2}} \right)^{\frac{1}{\eta}}, \quad (2.37)$$

which is a decreasing in productivity $z(\varphi_i)$.

Since

$$\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) = \exp \left\{ \frac{\hat{\theta}_{ip1}}{\eta} + \frac{1}{2} \left(\frac{\hat{\sigma}_{ip1}^2 + \sigma_\varepsilon^2}{\eta^2} \right) \right\},$$

where

$$\hat{\theta}_{ip1} = \frac{\left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip1} + \frac{\psi_{i1} \log s_{ip1}}{\sigma_\varepsilon^2}\right)}{\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} + \frac{\psi_{i1}}{\sigma_\varepsilon^2}\right)}, \text{ and } \hat{\sigma}_{ip1}^2 = \frac{1}{\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} + \frac{\psi_{i1}}{\sigma_\varepsilon^2}\right)},$$

Substituting the expressions for $\hat{\theta}_{ip1}$ and $\hat{\sigma}_{ip1}^2$, we can rewrite

$$\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) = \exp \left\{ \frac{1}{\eta} \left[\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} + \frac{\psi_{i1}}{\sigma_\varepsilon^2} \right)^{-1} \left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip1} + \frac{\psi_{i1} \log s_{ip1}}{\sigma_\varepsilon^2} + \frac{1}{2\eta} \right) + \frac{1}{2\eta} \sigma_\varepsilon^2 \right] \right\}.$$

Focusing on the terms involving ψ_{i1} , the derivative with respect to ψ_{i1} is proportional to

$$\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \right) \log s_{ip1} - \left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip1} + \frac{1}{2\eta} \right),$$

where the proportionality factor is strictly positive. Since $\log \theta_{ipt} = \log \theta_{ip} + \varepsilon_{ipt}$, where $\varepsilon_{ipt} \sim N(0, \sigma_\varepsilon^2)$, and $\log s_{ipt} = \log \theta_{ip} + v_{ipt}$, where $v_{ipt} \sim N(0, \frac{\sigma_\varepsilon^2}{\psi_{it}})$, taking expectations with respect to the noise terms yields

$$\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \right) \log \theta_{ip} - \left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip} + \frac{1}{2\eta} \right).$$

Therefore, for $\log \theta_{ip} > \theta_0 + \frac{\sigma_0^2}{2\eta}$, $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$ is increasing in ψ_{i1} , whereas for $\log \theta_{ip} < \theta_0 + \frac{\sigma_0^2}{2\eta}$, it is decreasing in ψ_{i1} . This implies that, for a given realization of θ_{ip1} , tenure $\tau = 1$, and productivity, firms with lower ψ_{i1} are more likely to underestimate $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$ when θ_{ip} is high. As a result, such firms are more likely to search for a new product ($\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) < \mathcal{B}_{i1}^{cs}$) or even drop the product ($\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) < \mathcal{B}_{i1}^{ds}$), even though the true θ_{ip} is high enough to justify continued operation without search.

Conversely, when θ_{ip} is low, firms with lower ψ_{i1} are more likely to overestimate $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$. In this case, they are more likely to continue operating the product, even though the true θ_{ip} would justify dropping it.

Furthermore, the likelihood that firms operate misaligned product lines for a given ψ_i gets higher if the firm has higher $z(\varphi_i)$. This is because, given all else equal, higher

$z(\varphi_i)$ increases the search threshold $\mathcal{B}_{i1}^{cs}$ and decreases the drop threshold $\mathcal{B}_{i1}^{ds}$ decreases in $z(\varphi_i)$, which widens the range of posterior mean $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$, over which firms both continue operating and search. This expanded region induces more firms—even those with relatively precise beliefs (high ψ_i)—to switch away from high- θ_{ip} products or retain low- θ_{ip} products. As a result, misallocation driven by learning frictions becomes more pronounced among high-productivity firms. $\square$

Since information-processing ability is closely linked to firm age, the model generates lifecycle patterns in product dynamics: when firms are young, they are less likely to retain products aligned with their expertise and more likely to retain misaligned products. These patterns suggest that firms make noisier product-portfolio decisions when they are young, due to limited information and lower information-processing ability. Furthermore, this is more pronounced for young firms with high potentials.

2.7 Social Planner's Problem

In this section, we solve the social planner's problem and derive further implications for the efficiency of the equilibrium. Given the symmetric stationary economy, we denote $\tilde{\mu}$ as the product-line distribution determined by the social planner and define the planner's problem as follows. The planner maximizes the period-by-period utility $\log \bar{C}$, where $\bar{C} = N^{\frac{\nu}{\nu-1}} C$, $C_p = C$, for all $p = 1, 2, \dots, N$, subject to the labor market clearing condition:

$$\max_{\{\tilde{c}(\xi), \tilde{\mathcal{B}}^{cs}(\xi), \tilde{\mathcal{B}}^{ds}(\xi), \tilde{\mathcal{M}}^e, \tilde{\varphi}^e, \tilde{m}_0, \tilde{z}_0\}} \log \left(N^{\frac{\nu}{\nu-1}} C \right)$$

subject to

$$C = \left[\int_{\mathcal{X}} \mathcal{B}(\xi) \tilde{c}(\xi)^{\frac{\eta-1}{\eta}} \tilde{\mu}(\xi) d\xi \right]^{\frac{\eta}{\eta-1}}$$

$$\bar{L} = N \int_{\mathcal{X}} \frac{\tilde{c}(\xi)}{z(\varphi)} \tilde{\mu}(\xi) d\xi + \mathcal{F}^o + \mathcal{F}^s + \mathcal{F}^e,$$

where $\mathcal{F}^o = N f^o \int_{\mathcal{X}} \tilde{\mu}(\xi) d\xi$, $\mathcal{F}^e = f^e \widetilde{\mathcal{M}}^e (1 + H(\underline{\varphi}^e))$, and $\mathcal{F}^s = N f^s \left(\int_{\mathcal{X}} \int_{\mathcal{X}} \mathbf{1}_{\mathcal{R}^{cs} \cup \mathcal{R}^{ds}}(\xi') g(\xi' | \xi) \tilde{\mu}(\xi) d\xi d\xi' + \int_{(a, \varphi)} \tilde{\mu}(\emptyset, \emptyset, a, \varphi) d(a, \varphi) \right)$.

Given this, we can characterize the planner's equilibrium in the following proposition.

Proposition 5. *Conditional on the cutoffs and entry mass, $\widetilde{\mathcal{B}}^{cs}(\xi)$, $\widetilde{\mathcal{B}}^{ds}(\xi)$, $\widetilde{\mathcal{M}}^e$, $\underline{\varphi}^e$, the planner's quantity decisions $\tilde{c}(\xi)$ for all active firms (with ξ) in the economy are the same as competitive equilibrium as follows:*

$$\tilde{c}(\xi) = \frac{\bar{L} - \mathcal{F}^o - \mathcal{F}^s - \mathcal{F}^e}{N} \left(\int_{\mathcal{X}} \mathcal{B}(\xi)^\eta z(\varphi)^{\eta-1} \tilde{\mu}(\xi) d\xi \right)^{-1} \mathcal{B}(\xi)^\eta z(\varphi)^\eta.$$

This implies that any stationary allocation that has the same entry and search/drop thresholds as the planner is efficient.

The social planner then chooses (i) the optimal switching rule, $\widetilde{\mathcal{B}}^{cs}(\xi)$, (ii) the optimal product-drop rule, $\widetilde{\mathcal{B}}^{ds}(\xi)$, and (iii) the optimal entry decisions, $\widetilde{\mathcal{M}}^e$ and $\underline{\varphi}^e$. Note that the first two cutoffs characterize product market entry and exit. The stationary product-line distribution $\tilde{\mu}$ is characterized by the following four equations based on such choices:

$$\begin{aligned} \tilde{\mu}(\hat{\theta}', \tau', a', \varphi) &= (1 - \delta) \int_{\mathbb{R}} \left[\mathbf{1}_{\widetilde{\mathcal{R}}^{ns}}(\hat{\theta}', \tau', a', \varphi) + (1 - x) \mathbf{1}_{\widetilde{\mathcal{R}}^{cs}}(\hat{\theta}', \tau', a', \varphi) \right] \\ &\quad \times g(\hat{\theta}', \tau', a', \varphi | \hat{\theta}, \tau' - 1, a' - 1, \varphi) \tilde{\mu}(\hat{\theta}, \tau' - 1, a' - 1, \varphi) d\hat{\theta} \\ \tilde{\mu}(\hat{\theta}^0, 0, a', \varphi) &= (1 - \delta) x \sum_{\tau'} \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}_{\widetilde{\mathcal{R}}^{cs} \cup \widetilde{\mathcal{R}}^{ds}}(\hat{\theta}', \tau', a', \varphi) \\ &\quad \times g(\hat{\theta}', \tau', a', \varphi | \hat{\theta}, \tau' - 1, a' - 1, \varphi) \tilde{\mu}(\hat{\theta}, \tau' - 1, a' - 1, \varphi) d\hat{\theta} d\hat{\theta}' \\ &\quad + (1 - \delta) x \tilde{\mu}(\emptyset, \emptyset, a' - 1, \varphi) + \frac{\widetilde{\mathcal{M}}^e}{N} h(\varphi) \mathbf{1}\{a' = 0\} \\ \mu(\emptyset, \emptyset, a', \varphi) &= (1 - \delta)(1 - x)(1 - p^e(a', \varphi)) \sum_{\tau'} \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}_{\widetilde{\mathcal{R}}^{ds}}(\hat{\theta}', \tau', a', \varphi) \\ &\quad \times g(\hat{\theta}', \tau', a', \varphi | \hat{\theta}, \tau' - 1, a' - 1, \varphi) \tilde{\mu}(\hat{\theta}, \tau' - 1, a' - 1, \varphi) d\hat{\theta} d\hat{\theta}' \end{aligned}$$

$$\begin{aligned}
& + (1 - \delta)(1 - x)(1 - p^e(a', \varphi))\tilde{\mu}(\emptyset, \emptyset, a' - 1, \varphi) \\
q^\emptyset(a', \varphi) = & \left[\tilde{\mu}(\emptyset, \emptyset, a' - 1, \varphi) + \sum_{\tau'} \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}_{\tilde{\mathcal{R}}^{ds}}(\hat{\theta}', \tau', a', \varphi) \right. \\
& \times g(\hat{\theta}', \tau', a', \varphi \mid \hat{\theta}, \tau' - 1, a' - 1, \varphi) \tilde{\mu}(\hat{\theta}, \tau' - 1, a' - 1, \varphi) d\hat{\theta} d\hat{\theta}' \left. \right] (1 - x).
\end{aligned}$$

Next, we derive the following corollary, which shows indicates room for government policies that can enhance welfare.

Corollary 1 (Constrained Inefficiency and Policy Implications). *If there exist lump-sum transfers to firms such that the resulting entry thresholds as well as search/drop thresholds coincide with those of the planner, $\tilde{\mathcal{B}}^{cs}(\xi)$, $\tilde{\mathcal{B}}^{ds}(\xi)$, $\tilde{\varphi}^e$, and thus, $\tilde{\mathcal{M}}^e$, then the competitive equilibrium is constrained efficient.*

This suggests that the extensive margins—both firm entry and exit in the economy and product-market entry and exit within incumbent firms—are not generally efficient in the competitive equilibrium. The source of this inefficiency is the interaction between learning and monopolistic competition.

When making entry or exit decisions, firms do not internalize either the benefits that additional product varieties generate for consumers or the negative effects that their entry imposes on incumbent firms through business stealing. In standard static models of monopolistic competition, these two forces exactly offset each other, implying efficient entry. In our model, however, learning creates an option value that is affected by entry and exit decisions, breaking this exact cancellation at both extensive margins.²²

As a consequence, the competitive equilibrium may exhibit inefficient firm entry and exit as well as inefficient product search and drop decisions. This creates a role for policies that target these extensive margins, such as entry subsidies, product-search subsidies, or bailout policies that affect firms' product drop or exit decisions.

²²This is consistent with [Arkolakis et al. \(2018\)](#) although their framework only features the firm entry and exit margin.

3 Data and Measurement

In this section, we empirically test the model’s predictions. Our main data source is the quinquennial Census of Manufacturers (CMF, henceforth) linked to the Longitudinal Business Database (LBD, henceforth) hosted by the U.S. Census Bureau. Our sample covers the following eight quinquennial census years: 1977, 1982, 1987, 1992, 1997, 2002, 2007, and 2012.

The CMF provides comprehensive information on all U.S. manufacturing establishments with one or more paid employees. The data collects establishment-level characteristics such as employment, payroll, worker hours, payroll supplements, cost of materials, selected operating expenses, value added, capital expenditures, inventories, energy consumption, and industry codes. It also provides the list of input materials and expenses used by each establishment. Furthermore, it contains product codes—either five-digit SIC or seven-digit NAICS codes—and the value of shipments for products manufactured by each establishment.²³ More details can be found from [Bernard et al. \(2010\)](#), [Kehrig et al. \(2011\)](#) and [Kehrig and Vincent \(2021\)](#).

The LBD tracks the universe of U.S. private non-farm business establishments and firms with at least one paid employee. It covers all sectors and geographic areas of the economy annually from 1976 onward. Establishments owned by a parent firm are grouped under a common firm identifier, enabling us to aggregate establishment-level activities to the firm level. The LBD contains basic information such as employment, payroll, industry codes, and location. These variables enable us to link to the CMF establishments and identify parent firms to construct firm-level variables such as firm size, age, productivity, employment growth rates, and entry/exit. We refer to [Jarmin and Miranda \(2002\)](#), and [Chow et al. \(2021\)](#) for further details.

Industry-level variables are adopted from the NBER CES Manufacturing Industry Database, assembled by [Becker et al. \(2013\)](#). An industry is defined by the four-digit

²³For example, in the Nonferrous Wiredrawing and Insulating industry (SIC 3357), there are thirteen products, including Aluminum Wire (33571), Copper Wire (33572), Telephone Wire (3357B), and Fiber Optic Cable (33579).

SIC (or the six-digit NAICS after 1997) code. The NBER-CES Manufacturing Industry Database contains annual industrial data for the U.S. manufacturing sector from 1958 through 2018, sourced from the U.S. Census Bureau, the Bureau of Economic Analysis, and the Bureau of Labor Statistics. The data covers industry-level information on price deflators, payroll, employment size, number of workers, total value of shipments, value added, various input costs and expenditures, and productivity (TFP estimates).

In addition, we use the Management and Organizational Practices Survey (MOPS), a plant-level survey of management practices and organizational structure in the U.S. manufacturing sector conducted in 2010, 2015, and 2021.²⁴ We construct three measures of managerial ability from the survey responses. The first is the average structured management score developed by Bloom et al. (2019). The other two are based on responses to questions related to data-driven decision making, capturing data availability and the intensity of data use, respectively. We use the plant-level survey responses and aggregate them to the firm level using value-of-shipments weights.

3.1 Product Definition and Characteristics

We define a product by a five-digit SIC code for the entire census years.²⁵ Note that in each Census year, there are two versions of product codes: one based on the previous Census year’s classification (“as collected”) and another based on the current Census year’s classification. For 1977–1997, the “current” classification is based on the 1987 SIC system. Starting from 2002, product codes based on the NAICS classification are used. Importantly, in the 1997 CMF file, both the 1997 NAICS-based product code and the 1987 SIC-based product code appears, allowing us to construct a time-consistent product classification based on the 1987 SIC.

The bridge construction from the 1987 SIC to any version of NAICS follows a rolling

²⁴See Bloom et al. (2019) for additional details.

²⁵This is based on product trailer information provided by the Census for the U.S. manufacturing firms in the CMF. We extract the information at the SIC5 or NAICS7 level for the pre-1997 and post-1997 Census years, respectively.

approach. Starting with the 1987 SIC-to-1997 NAICS concordance, we then link the 1997 NAICS to the 2002 NAICS using the 2002 Census bridge file, yielding a concordance between the 1997 SIC and 2002 NAICS. We repeat this process for the 2007 and 2012 Censuses, ultimately generating a 1987 SIC to 2002, 2007, and 2012 NAICS concordance. To maximize within-firm consistency across years from 2002 to 2012, we first match each firm-product pair in a given year to its corresponding pair in the previous year. For unmatched cases, we use the bridge file constructed from the previous year to map the NAICS product code to its corresponding 1987 SIC product code. If a NAICS product code maps to multiple 1987 SIC codes, we select the one most frequently matched to that NAICS code.

The CMF reports product-level sales (product value of shipment) and physical quantities shipped. Given the definition of a product, we limit our analyses to observations that have a positive, nonzero value of shipments. Following literature, we drop “administrative record,” which are very small establishments that are not required to report the product-level value of shipments. We also exclude observations outside the “tabbed sample,” which is the sample used by the Census Bureau for official tabulations. We use the product value of shipment at the product-establishment level and aggregate it up to the product-firm level using LBD firm identifiers. We normalize the product value of shipments using the NBER-CES industry-level price indices (in 1997 U.S. dollars). For each product-firm pair, we measure tenure as the number of years the firm manufactures the product and appears in the record. To improve accuracy, we supplement our data with product-firm information from the Annual Survey of Manufacturers (ASM).

Lastly, for each firm in year t , we flag “added products” as those that were not present in the firm’s product portfolio before year t but appear by $t + 5$. In a similar fashion, we flag “dropped products” within each firm in year t as those that are present in the firm’s product portfolio up to year t but no longer appear from $t + 5$. This enables us to identify added and dropped products from 1982 to 2007 in our sample.²⁶ For

²⁶We use the 1977 data as a buffer to more accurately flag product additions and drops and compute product-firm tenure.

each added and dropped product, we indicate whether they are initial additions or re-entries, and whether drops are temporary or permanent. Our baseline analysis focuses on initial additions and permanent drops.

3.2 Firm Characteristics and Identifiers

We use firm age variables constructed by the Census using the method in [Haltiwanger et al. \(2013\)](#). Specifically, firm age is defined as the age of the oldest establishment owned by the firm when it is first observed in the data. We indicate young firms as those younger than or equal to age five. Note that because the LBD begins in 1976, all firms are assigned age zero in that year. To reduce misclassification of young firms, we use the 1972 CMF to adjust firm ages accordingly.²⁷ Firm size is measured using either total employment or total value of shipments.

One limitation of the LBD is the lack of longitudinally consistent firm identifiers. Although the redesigned LBD has a new firm identifier that links firms across time by correcting previous firm identifiers that are recycled in the old LBD, as described in [Chow et al. \(2021\)](#), it still has limitations.²⁸ However, longitudinal consistency of firm identifiers is necessary for our analysis to track firms' history of product portfolios. Therefore, we construct alternative versions of longitudinal firm identifiers from the LBD following [Dent et al. \(2018\)](#) and use them to assess the robustness of our analysis results. Henceforth, we use the term "firm identifier" to refer to the firm identifiers constructed by the Census Bureau (lbfid). All nominal values are converted to 1997 U.S. dollars using industry-level deflators for the relevant variables.

²⁷We use Finished Goods Inventory Beginning (FIB) from the 1972 CMF and add 4 to the firm age of LBD firms that own 1972 CMF establishments if FIB equals zero, and 5 years if FIB is greater than zero.

²⁸The new firm identifiers haven't still resolved firm reorganization issue. See more discussion in [Chow et al. \(2021\)](#).

Table 1: Share of Product Switching Firms

variables	Including entry and exit			Continuers only		
	All	Age 0-5	Age 6 +	All	Age 0-5	Age 6 +
	Product adding only					
Share of firms	0.094	0.100	0.093	0.128	0.130	0.128
Output share	0.058	0.098	0.057	0.067	0.139	0.066
	Product dropping only					
Share of firms	0.398	0.329	0.407	0.182	0.143	0.187
Output share	0.282	0.404	0.280	0.169	0.146	0.169
	Product churning					
Share of firms	0.248	0.310	0.239	0.336	0.394	0.327
Output share	0.582	0.284	0.588	0.674	0.409	0.678

Notes: Firms in the manufacturing sector from 1982 to 2007.

4 Empirical Models and Main Results

In this section, we investigate firms' product-switching decisions and examine how the decisions differ for young firms. The design of our empirical strategy is discussed in the following subsections.

4.1 Prevalence of Product Switching Firms

Table 1 reports the share of product-switching firms and their output shares by age group, with and without entry and exit. Since firm entry and exit can overstate firm-level product additions and drops, the last three columns report shares for continuing firms only. Focusing on continuing firms, we find that 65% of firms in the pooled sample switch products at least once every five years, and they account for 91% of total output. Thus, product switching is a prevalent behavior, and firms that engage in it account for a substantial share of economic activity, as emphasized by [Bernard et al. \(2010\)](#).

Table 1 also shows that product switching often involves simultaneous add and drop of products (churning). Approximately 52% of product-switching firms (34% out of 65%) both add and drop products, while the remainder either add or drop, but not both. The life-cycle pattern of product switching exhibits distinct dynamics. In particular,

Table 2: Share of Multi-Product Firms

Classification	Share of firms			Output share		
	All	Age 0-5	Age 6 +	All	Age 0-5	Age 6 +
sic5	0.450	0.432	0.452	0.891	0.615	0.896
sic4	0.332	0.301	0.336	0.838	0.451	0.845
sic2	0.183	0.153	0.187	0.702	0.267	0.711

Notes: Firms in the manufacturing sector from 1982 to 2007.

product churning is more pervasive among young firms, while the share of drop-only cases is lower compared to their more mature counterparts.

Table 2 additionally reports the share of multi-product firms under alternative definitions of a product: the baseline five-digit SIC product code, the four-digit SIC industry code, and the two-digit SIC sector code. Consistent with existing findings, multi-product firms comprise slightly less than half of all firms (45%) but account for 89% of total output. These multi-product firms also operate across multiple industries and sectors. Multi-product firms are also prevalent among young firms, comprising 43% of them and generating 62% of their output.

In contrast, while single-product firms make up 55% of all firms, they account for only 11% of total output. Moreover, firms that produce a single product and never switch their product throughout the sample period represent 14.8% of firms but contribute just 2.8% of total output.

Lastly, Table 3 reports the means and standard deviations of the number of products firms produce, add, and drop, based on the full sample including single-product firms. On average, firms produce 2.2 products, add 1.2 products, and drop 0.9 products over five years.²⁹ Although young firms produce fewer products on average (1.6 vs. 2.3), they tend to add more products (1.6 vs. 1.2) while dropping fewer products (0.8 vs. 1.0) compared to mature firms.

²⁹Bernard et al. (2010) report that the average number of products firms produce rises to 3.5 when single-product firms are excluded.

Table 3: Number of Products Produced/Added/Dropped

	Including entry and exit			Continuers only		
	All	Age 0-5	Age 6 +	All	Age 0-5	Age 6 +
Number of products firms produce						
Mean	2.189	1.877	2.233	2.192	1.645	2.269
Stdev.	(2.896)	(1.509)	(3.041)	(3.153)	(1.377)	(3.321)
Number of products firms add						
Mean	1.27	1.831	1.189	1.244	1.605	1.194
Stdev.	(1.831)	(1.518)	(1.858)	(1.880)	(1.374)	(1.935)
Number of products firms drop						
Mean	1.273	1.218	1.281	0.941	0.790	0.963
Stdev.	(1.981)	(1.482)	(2.043)	(1.841)	(1.101)	(1.922)

Notes: Firms in the manufacturing sector from 1982 to 2007.

4.2 Product-Specific Expertise Estimation

We estimate product-specific expertise of firms and examine the relationship between firms' product-switching decisions and resource reallocation. In the CMF, the only performance measure available at the product-firm level is the product value of shipments (PV).³⁰ However, using PV directly to assess whether a firm is relatively better at producing one product over another presents several issues. The level of PV may reflect not only firm-specific capabilities but also factors such as the duration of production (e.g., life-cycle pattern of product as in [Argente et al. \(2024\)](#) or customer capital accumulation) and transitory product-level shocks (e.g., market fluctuations).

To address these concerns, we estimate a time-invariant measure of product-firm performance by taking out factors that influence the level of PV other than the firm's product-specific ability. We refer to this measure as firms' product-specific expertise, or the match (fit) between a firm and a product, which captures the firm's capability to produce a given product. We estimate the following baseline model, similar to [Bonhomme et al. \(2019\)](#):

$$y_{ipjt} = \theta_{ip} + X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_{pt} + \varepsilon_{ipjt}, \quad (4.38)$$

³⁰While quantity data are also available, they cover only a limited set of products.

y_{ipjt} is the log real PV for firm i (in industry j) and product p in census year t . θ_{ip} , a firm-product fixed effect, is our main interest. This contains time-invariant characteristics associated with the firm-product match that determine the performance of the product within the firm and capture the firm's fundamental expertise in the product.

We include a rich set of controls and fixed effects to account for sources affecting the firm-product-specific value of shipments, unrelated to the firm-product-specific unobservable characteristics. For example, when firms enter new product markets, it typically takes time to build a customer base, which can affect shipment values independently of its expertise. To control for this effect, we include product-firm-specific tenure as of year t and its square. Furthermore, we include a vector of firm-level time-varying characteristics (X_{ijt}), which includes firm size (log employment), firm age, total assets, number of products, number of establishments, cost of advertisements, cost of software, skill and capital intensity (log ratio of non-production workers to total employment and capital to total employment), productivity, and firm-level inputs (log of production workers, equipment capital stock, structural capital stock, material cost, and energy costs).

Lastly, δ_{pt} is a product-year fixed effect to control for any potential factors affecting the value of shipment for firm i 's product p , attributed to time-varying technology or demand structure that differs across each product market. Related to this, several product-specific properties (skill or capital intensity) can also affect the total value of shipment regardless of the firm-product's fundamental expertise. As robustness checks, we further explore alternative specifications to estimate the product-firm match. More discussion can be found in the Appendix.

Note that as we control for factors specific to product markets and firms each year, our measure for the product-firm match, θ_{ip} , can be compared across different product-firm pairs across different years. We rescale the quality measure to make it normalized and range in $[0,1]$. Let q_{ip} denote this. In particular, we use the following formula to

rescale $\hat{\theta}_{ip}$:

$$q_{ip} \equiv \frac{\hat{\theta}_{ip} - \underline{\theta}_{ip}}{\bar{\theta}_{ip} - \underline{\theta}_{ip}}, \quad (4.39)$$

where $\underline{\theta}_{ip}$ and $\bar{\theta}_{ip}$ are the minimum and maximum value of the quality estimates, respectively (e.g. $\hat{\theta}_{ip} \in [\underline{\theta}_{ip}, \bar{\theta}_{ip}]$).³¹ We estimate θ_{ip} using the CMF for the years 1977 to 2012, complemented with the ASM data from 1976 to 2016.

4.3 Product Dropping

In this subsection, we examine firms' product-dropping decisions and analyze how product-dropping patterns differ between young and mature firms.

We begin by examining the types of products that firms choose to drop. Specifically, we test whether firms are more likely to drop products with lower performance—measured either by contemporaneous performance or by long-run performance—relative to products that perform better. This prediction is based on the premise that the quality of a product-firm match is not directly observable and can only be learned over time through costly experimentation and information acquisition. Such learning frictions are likely to be more pronounced for younger firms, which possess less information about their own capabilities and have accumulated less experience in product markets.

To see this, we construct a dummy variable, $\mathcal{I}_{ipjt}^{drop}$, to indicate a product being dropped by the beginning of the next period. $\mathcal{I}_{ipjt}^{drop}$ is equal to one if firm i in industry j drops product p between period t and $t + 5$ (e.g., if product p appears in firm i 's portfolio in t but not $t + 5$). We regress it on the product-firm performance measures to see how firms' product drop depends on the match between product p and firm i :

$$\mathcal{I}_{ipjt}^{drop} = \alpha + \beta Q_{ipt} + X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_i + \delta_{pt} + \varepsilon_{ipjt}. \quad (4.40)$$

³¹As a robustness check, we also use alternative normalization methods, including standardization relative to the industry-year (or product-year) cross-sectional distribution or percentile rankings within industry-year (or product-year) cells. The results are robust to these alternative measures.

Table 4: Probability of Product Drop

	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
$\log(PV_{pit})$	-0.052*** (0.002)			-0.052*** (0.002)	
θ_{ip}		-1.074*** (0.021)			-1.081*** (0.022)
Young Firm $_{it}$			-0.075*** (0.003)	-0.141*** (0.007)	-0.368*** (0.022)
$\log(PV_{pit}) \times \text{Young}_{it}$				0.010*** (0.001)	
$\theta_{ip} \times \text{Young}_{it}$					0.497*** (0.043)
Observations	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000
Fixed effects	pt, i	pt, i	pt, i	pt, i	pt, i
Controls	Full	Full	Full	Full	Full

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Q_{ipt} is either the log of real PV or q_{ip} , the long-run performance (θ_{ip} estimates in 4.39) associated with the pair of product p and firm i . X_{ipjt} is a vector of time-varying firm-product characteristics (log firm-product specific tenure and square, and log value of shipment if q_{ip} is used for Q_{ipt}). X_{ijt} is a vector of firm controls (the same set of controls used in q_{ip} estimation). δ_{pt} is a product-year fixed effect, and δ_i is a firm fixed effect.

The first two columns of Table 4 present results from specifications that include the full set of control variables.³² As shown in column 1 of Table 4, firms are, on average, more likely to drop products that generate lower revenue. Column 2 further shows that this pattern holds for our time-invariant firm-product performance measure: firms tend to drop products with lower long-run performance, relative to better-matched ones. These results suggest that, on average, firms are able to identify which products are better matches and drop those that are poorly matched. Dropping poorly matched products would free up the resources and potentially enable firms to reallocate these

³²Full tables are available in Appendix B.1.

resources toward their better use.

4.3.1 Product Dropping and Firm Lifecycle

Next, we test how product-dropping behavior varies by firm age. First, we estimate the following regression model:

$$\mathcal{I}_{ipjt}^{drop} = \alpha + \beta \mathcal{I}_{ijt}^{young} + X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_i + \delta_{pt} + \varepsilon_{ipjt}, \quad (4.41)$$

where $\mathcal{I}_{ijt}^{young}$ is a dummy variable equal to one if a firm is with age less than or equal to five in year t .

One thing to note is that single-product firms could choose not to drop their product even though they know that the product they produce is a poor match for them. This is because dropping the product would force them to shut down their business, as it is their only product. Because young firms are more likely single-product firms, including the number of products firms produce as a firm-level control is particularly critical in this regression model.³³ Financial frictions can also matter. The product-switching behavior of financially constrained firms may differ from that of unconstrained firms, as the former face greater difficulty in financing new projects to add products. One way to mitigate this concern is to include a proxy for firms' ability to raise funds. The closest available variable in the CMF is total assets.³⁴ We include these variables, as well as other firm controls as before. The third column of Table 4 presents the result. This shows that on average, young firms drop products less frequently than mature firms.³⁵

To further explore the type of products dropped by young firms, we interact the two firm-product performance measures in equation (4.40) with the young firm dummy as

³³As shown in Table 2, young firms are approximately 4% less likely to be multi-product firms compared to mature firms (43% vs. 45%).

³⁴We are in the process of exploring another way to address financial frictions, such as industry-level differences in external financing arising from the relative importance of intangible versus physical capital.

³⁵Full tables are available in Appendix B.1, including those with and without firm-level controls.

follows:

$$\mathcal{I}_{ipjt}^{drop} = \alpha + \beta_1 Q_{ipt} + \beta_2 \mathcal{I}_{ijt}^{young} + \beta_3 Q_{ipt} \times \mathcal{I}_{ijt}^{young} + X_{ipjt} \gamma_1 + X_{ijt} \gamma_2 + \delta_i + \delta_{pt} + \varepsilon_{ipjt}. \quad (4.42)$$

Columns 4 and 5 of Table 4 lay out the results.³⁶ In particular, the third row (the coefficient on the young firm dummy) indicates that with the level of revenue controlled, young firms drop products even less frequently than their mature counterparts on average.

However, as shown in the fourth and fifth rows, young firms are more likely to drop products in which they perform well while dropping fewer products with poor performance, relative to mature firms. In other words, young firms tend to drop products they should have retained while retaining products they should have dropped. Overall, the stickiness with poorly-performing products is the dominant factor determining the average product-dropping behavior of young firms.

These results suggests that young firms' limited information for assessing product match may constrain their ability to keep better-matched products while dropping poorly matched ones at the early stage of their life-cycle. As a result, young firms may struggle to accurately assess whether a product is a good match for them and find the right set of products to operate. Such a restriction can hinder the potential growth of young firms, especially in environments with greater noise and less accurate information for firms to learn, such as during recessions.³⁷

Furthermore, we examine how young firms add and retain products. To see this, we define a product as being added to a firm's portfolio in year t if it is absent in the firm's portfolio in t but appears by the next census year $t + 5$. Let $\mathcal{I}_{ijt}^{add}$ denote a dummy variable equal to one if firm i in industry j adds at least one product between t and

³⁶Full tables are available in Appendix B.1.

³⁷We also find consistent results over the business cycle, where such patterns of product switching for young firms are more pronounced in recessions. Results are available on request. The results are also robust when replacing the young firm dummy with firm age, as presented in Appendix Table A14.

Table 5: Product Add and Retention

	$Add_{t:t+5}$	$Add_{t:t+5}^{high}$
Young Firm $_{it}$	0.148*** (0.005)	-0.112*** (0.017)
Observations	682,000	66,000
Fixed effects	i, jt	i, jt
Controls	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

$t + 5$. Additionally, to investigate the probability of adding (and retaining) a better-performing product, we define $\mathcal{I}_{ijt}^{add,high}$ as a dummy variable equal to one if the average long-run performance of products newly added and retained between t and $t + 5$ is higher than the average long-run performance of the products firms produce in t .

With the two variables, we estimate the following regression with other factors controlled as before:

$$\mathcal{I}_{ijt}^{add} \text{ (or } \mathcal{I}_{ijt}^{add,high} \text{)} = \alpha + \beta \mathcal{I}_{ijt}^{young} + X_{ijt}\gamma + \delta_i + \delta_{jt} + \varepsilon_{ijt}, \quad (4.43)$$

where δ_i is a firm fixed effect, and δ_{jt} is an industry-year fixed effect. The remaining firm-level controls are identical to those used previously.

The results are shown in Table 5. The first column shows that young firms are more likely to add new products on average. However, as presented in the second column, young firms are less likely to add and retain better-performing products relative to their mature counterparts.³⁸ This also suggests that young firms face greater difficulty in identifying and retaining the right products. While they actively experiment by adding new products, many of these additions do not align well with their underlying expertise.

Combined with the previous findings, this indicates that firms learn about their exper-

³⁸Full tables are available in Appendix B.2.

Table 6: Probability of Product Drop

	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-1.055*** (0.041)	-1.127*** (0.033)	-1.119*** (0.035)
Management $_{it}$	0.465*** (0.041)	0.406*** (0.033)	0.427*** (0.036)
$\theta_{ip} \times \text{Management}_{it}$	-1.036*** (0.082)	-0.832*** (0.062)	-0.912*** (0.068)
Observations	1,622,000	1,622,000	1,622,000
Fixed effects	pt, i	pt, i	pt, i
Controls	Full	Full	Full
Management type	General	Data availability	Data use

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the five-digit SIC product level are reported in parentheses. Columns (1) uses the overall management index as in Bloom et al. (2019), column (2) uses the data-availability measure (Q27), and columns (3) uses the data use intensity measure (Q28). Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

tise while operating, enabling them to reoptimize their product portfolios by retaining products that are better matched to their capabilities while dropping the ones that are not. However, young firms appear to face challenges in this process.

4.3.2 Product Dropping and Firm Managerial Ability

Next, we investigate how product dropping behavior varies by firm managerial ability. To see this, we estimate the following regression model:

$$\mathcal{I}_{ipjt}^{drop} = \alpha + \beta_1 \theta_{ip} + \beta_2 \text{Management}_{ijt} + \beta_3 \theta_{ip} \times \text{Management}_{ijt} + X_{ipjt} \gamma_1 + X_{ijt} \gamma_2 + \delta_i + \delta_{pt} + \varepsilon_{ipjt}, \quad (4.44)$$

where Management_{ijt} is the managerial ability measure constructed for firm i at t using the three types: general score (Bloom et al., 2019), data availability, and data use intensity. All else variables are defined as the same as before.

Table 6 shows the results, where firms with better management are more likely to

Table 7: Product Add/Drop and Firm Performance

	ΔV_{add}	ΔEmp	ΔTFP	ΔLP
$Add_{t:t+5}$	0.830*** (0.019)	0.802*** (0.019)	0.735*** (0.019)	0.762*** (0.019)
$Drop_{t:t+5}$	-0.754*** (0.019)	-0.731*** (0.018)	-0.686*** (0.017)	-0.713*** (0.018)
Observations	682,000	682,000	648,000	648,000
Fixed effects	i, jt	i, jt	i, jt	i, jt
Controls	Full	Full	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

drop poorly-performing products. This holds across the three measures of managerial ability.³⁹

4.4 Product Switching and Firm Performance

Lastly, we further examine whether firms' product-adding or dropping behavior affects firm performance.

4.4.1 The Impact of Product Adding and Dropping

We begin by using the following regression model to estimate the relationship between the decision to add or drop a product and firm performance:

$$\Delta y_{ijt} = \alpha + \beta_1 \mathcal{I}_{ijt}^{drop} + \beta_2 \mathcal{I}_{ijt}^{add} + X_{ijt} \gamma + \delta_i + \delta_{jt} + \varepsilon_{ijt}, \quad (4.45)$$

where Δy_{ijt} is the log difference of i) real value added, ii) total employment, iii) total factor productivity, or iv) labor productivity between t and $t + 5$.⁴⁰

³⁹Full tables are available in Appendix B.1.

⁴⁰Value added is Total Value of Shipments + Final Inventory Investment + Work-in-Progress Goods - Resales - Material Inputs - Energy Expenditures. Labor productivity is the Real value of shipments/Total employment. We use the Census TFP beta 8 available in the CMF.

Table 8: Firm-Level Output and Long-run Performance Growth Decomposition

	Output			Long-run Performance (θ_{ip})		
	all	age 0-5	age 6 +	all	age 0-5	age 6 +
Total changes	0.112	0.662	0.104	0.017	0.130	0.016
Within share	0.543	0.444	0.545	0	0	0
Between share	0.008	0.002	0.009	0.318	0.126	0.320
Cross share	0.009	-0.005	0.010	0	0	0
Addition share	0.424	0.554	0.422	0.574	0.853	0.572
Drop share	0.015	0.005	0.015	0.107	0.022	0.108

Notes: Firms in the manufacturing sector from 1982 to 2007.

Table 7 presents the result, indicating that product dropping is negatively associated with subsequent growth across the four performance measures, while product adding is positively correlated with them.⁴¹ When considering product switching—defined as the combination of adding and dropping products—it is positively associated with subsequent changes in all four performance measures.⁴²

4.4.2 Decomposition of Firm-Level Growth in Output and Product Expertise

To understand the importance of within-firm reallocation of resources through product switching to firm growth further, we conduct an accounting decomposition of output and long-run performance growth at the firm-product level. This decomposition allows us to distinguish between the effects of changes in the share of output across products and output and long-run performance growth within each firm-product pair.

Define within-firm output share of product p at t for firm i as $\omega_{ipt} = \frac{PV_{ipt}}{\sum_{p \in \mathcal{P}_{it}} PV_{ipt}}$, where PV_{ipt} denotes the real product value of shipments of product p by firm i at t , and $\mathcal{P}_{it}$ is the set of products in firm i 's product portfolio at t . Let y_{ipt} denote either the log of real product value of shipments, $\log(PV_{ipt})$, or the firm-product long-run performance, θ_{ip} . We compute the firm-level average output or long-run performance

⁴¹Table A9 in the Appendix reports the full set of results.

⁴²These results are consistent with previous findings by Bernard et al. (2010) and Argente et al. (2018).

as $y_{it} = \sum_{p \in \mathcal{P}_{it}} \omega_{ipt} y_{ipt}$.

We then decompose the growth in firm-level output or long-run performance as follows:

$$\begin{aligned}
\Delta y_{it} = & \underbrace{\sum_{p \in \mathcal{C}_{it}} \omega_{pit-5} \Delta y_{ipt}}_{\text{within term}} + \underbrace{\sum_{p \in \mathcal{C}_{it}} (y_{pit-5} - y_{it-5}) \Delta \omega_{ipt}}_{\text{between term}} + \underbrace{\sum_{p \in \mathcal{C}_{it}} \Delta \omega_{ipt} \Delta y_{ipt}}_{\text{cross term}} \\
& + \underbrace{\sum_{p \in \mathcal{N}_{it}} \omega_{ipt} (y_{ipt} - y_{it-5})}_{\text{addition term}} - \underbrace{\sum_{p \in \mathcal{X}_{it}} \omega_{pit-5} (y_{pit-5} - y_{it-5})}_{\text{drop term}}. \tag{4.46}
\end{aligned}$$

$\Delta X_{.t} = X_{.t} - X_{.t-5}$ is the change in any variable $X_{.t}$ between $t-5$ and t , $\mathcal{C}_{it}$ is the set of continuing products that firm i produces in both $t-5$ and t , $\mathcal{N}_{it}$ is the set of products newly added by firm i in t , and $\mathcal{X}_{it}$ is the set of products dropped by firm i in t .

Table 8 presents the decomposition results for both output (real product value of shipments) and match performance (θ_{ip}). We normalize each term by the total change to quantify its contribution to overall growth. We refer to these normalized values as shares. The first three columns show that output growth is primarily driven by within-product growth (within share) and new product addition (addition term). For young firms, however, new product addition plays a more prominent role than within-product growth. In fact, the combined contribution of product-switching components, addition and drop shares, is 28% higher for young firms than for mature firms ($0.554 + 0.005$ vs. $0.422 + 0.015$).

Taken together, these results provide suggestive evidence that firms foster growth by reallocating resources through product switching, effectively improving the match outcome and performance. However, young firms struggle in this process, often devoting time and resources to discovering their areas of expertise by experimenting with a wider range of products.

4.5 Robustness Tests

We conduct several robustness checks for the regression results. Results are shown in Appendix B.3 and Appendix B.4.

First, the pair fixed-effects specification may still be subject to selection concerns. In particular, young firms tend to have fewer observations within a given firm–product pair, which could affect the estimation. To address this issue, we re-estimate the regressions using only the first two observations for each firm–product pair and find robust results. We also address potential sampling bias by estimating the regressions separately over rolling windows of 5, 10, and 15 years. We confirm that the results are robust across all specifications.

In addition, we explore alternative specifications for estimating product–firm pair fixed effects. Specifically, we vary the set of firm-level controls included in the estimation and also consider a specification without any firm controls. As a particularly stringent alternative, we replace the firm-level controls with firm-year fixed effects, thereby absorbing all time-varying firm-level heterogeneity. We also implement an AKM-style specification (Abowd et al., 1999), and construct the product–firm effect as the time average of the residuals.⁴³ Across all specifications, the main results remain robust.

Second, for regressions that rely on estimated fixed effects, we adjust standard errors with bootstrapping. The results remain unchanged.

Third, in the product-dropping analysis, firm exit may affect the estimates through several channels. To address this concern, we re-estimate the models after excluding all firms that subsequently exit. The results remain robust.

Finally, our findings are robust to alternative measures of product proximity. In addition to our baseline measure, we construct proximity measures based on the similarity of material input usage across products and on industry classifications. The results are consistent.

⁴³See more discussion in Appendix B.5.

5 Quantitative Analysis

TBA

6 Concluding Remarks

In this paper, we develop a model of product switching over the firm lifecycle in which firms learn about product-specific expertise and adjust their product portfolios accordingly. Firms' ability to process information improves over time, which leads to imperfect learning early in the lifecycle. The learning process generates systematic differences in beliefs and shapes product-switching decision of firms and within-firm resource allocation across products. As a result, firms adjust their product portfolios more slowly toward products in which they can perform well when they are young, with implications for firm growth and performance.

We take the model to a comprehensive administrative dataset that tracks U.S. manufacturing output at the firm–product level and find empirical evidence supporting the model. Firms tend to drop products with low performance, but this pattern is weaker among young firms. In particular, young firms are more likely to retain low-performing products and to drop high-performing products. This pattern is observed for firms with lower managerial ability as well.

The model provides a unified interpretation of these patterns. Firms with lower information-processing ability—such as young firms—place greater weight on prior beliefs and rely on noisier signals, making them more likely to retain poorly performing products and drop better-performing ones. This mechanism links lifecycle patterns in product switching to differences in firm performance and growth.

These findings have broader implications for resource allocation. With product-switching decisions of firms, the learning friction creates a within-firm margin of misallocation that complements the standard cross-firm selection channel in shaping aggregate productivity. This perspective enables us to assess the aggregate importance of

firms' product-portfolio adjustments—a previously unexplored extensive margin—and to analyze the effectiveness of policies aimed at influencing product search, product reallocation, and product exit decisions.

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Appendix

A Theory

A.1 Proofs

Proof of Lemma 1, Belief Formation

Proof. The first derivative of $\hat{\theta}_{ipt}$ with respect to ψ_{it} is given by

$$\frac{\partial \hat{\theta}_{ipt}}{\partial \psi_{it}} = \frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2} \left[\log s_{ipt} \left(\frac{1}{\sigma_0^2} + \frac{\tau}{\sigma_\varepsilon^2} + \frac{\sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} \right) - \left(\frac{\theta_0}{\sigma_0^2} + \frac{\sum_{\tilde{t}} \log \theta_{ip\tilde{t}}}{\sigma_\varepsilon^2} + \frac{\sum_{\tilde{t}} \psi_{i\tilde{t}} \log s_{ip\tilde{t}}}{\sigma_\varepsilon^2} \right) \right].$$

Since $\log \theta_{ipt} = \log \theta_{ip} + \varepsilon_{ipt}$, where $\varepsilon_{ipt} \sim N(0, \sigma_\varepsilon^2)$, and $\log s_{ipt} = \log \theta_{ip} + v_{ipt}$, where $v_{ipt} \sim N(0, \frac{\sigma_\varepsilon^2}{\psi_{it}})$, taking expectations with respect to the noise terms yields

$$\mathbb{E}_{\varepsilon, v} \left(\frac{\partial \hat{\theta}_{ipt}}{\partial \psi_{it}} \right) = \frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2 \sigma_0^2} (\log \theta_{ip} - \theta_0).$$

Therefore, on average, when $\log \theta_{ip} > \theta_0$, a higher ψ_{it} raises the posterior mean, implying that firms with lower ψ_{it} underestimate $\hat{\theta}_{ipt}$. Conversely, when $\log \theta_{ip} < \theta_0$, a higher ψ_{it} lowers the posterior mean, so firms with lower ψ_{it} overestimate $\hat{\theta}_{ipt}$. Moreover,

$$\frac{\partial \hat{\sigma}_{ipt}^2}{\partial \psi_{it}} = -\frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2} < 0.$$

Therefore, firms with lower information-processing ability have a higher posterior variance. □

Proof of Lemma 5, Independence of Product Line Values

Proof. The value of product line k is determined by flow profits, the switching decision, and the firm's exit decision. Flow profits depend only on $(\hat{\theta}_{ipkt}, \hat{\sigma}_{ipkt}, \theta_{ipkt}, \varphi_i)$, and the

evolution of $(\hat{\theta}_{ip_k t}, \hat{\sigma}_{ip_k t}, \theta_{ip_k t})$ depends only on firm-level states $(a_{it}, \psi_{it}, \varphi_i)$ and product-specific factors.

When $\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) < \mathcal{B}_{it}^d(\hat{\theta}_{ipt}, \tau_{ipt})$, the continuation value depends on whether the firm exits or continues to search. By assumption, both outcomes yield the same continuation value $\beta V^{ss}(\mathbf{0}; a_{it+1}, \psi_{it+1}, \varphi_i)$. Therefore, although the probability of exit depends on the evolution of match qualities in other product lines, this dependence does not affect the value of product line k , since exit and continuation are payoff-equivalent.

It follows that the value of product line k depends only on $(\hat{\theta}_{ip_k t}, \tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i)$ and not on Θ_{it} . This establishes independence across product lines. $\square$

Proof of Proposition 3, Learning Frictions and Output Wedges

Proof. Recall that the optimal output for product p by firm i with productivity φ_i , given beliefs $(\hat{\theta}_{ipt-1}, \tau_{ipt-1})$ in period t , is

$$y_{ipt}^* = \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt}^\eta C_{pt} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta z(\varphi_i)^\eta w_t^{-\eta},$$

where $\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) = \exp \left\{ \frac{\hat{\theta}_{ipt-1}}{\eta} + \frac{1}{2} \left(\frac{\hat{\sigma}_{ipt-1}^2 + \sigma_\varepsilon^2}{\eta^2} \right) \right\}$, the posterior variance is given by $\hat{\sigma}_{ipt-1}^2 = \left(\frac{1}{\sigma_0^2} + \frac{\tau_{ipt-1}}{\sigma_\varepsilon^2} + \frac{\sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} \right)^{-1}$, and the posterior mean can be written as

$$\hat{\theta}_{ipt-1} = \hat{\sigma}_{ipt-1}^2 \left[\frac{\theta_0}{\sigma_0^2} + \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}} \log \theta_{ip}}{\sigma_\varepsilon^2} + \frac{1}{\sigma_\varepsilon^2} \left(\sum_{\tilde{t}} \varepsilon_{ip\tilde{t}} + \sum_{\tilde{t}} \psi_{i\tilde{t}} v_{ip\tilde{t}} \right) \right].$$

Taking expectations, we obtain the average value of $\left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta$ as

$$\begin{aligned} \mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta &= \exp \left[\hat{\sigma}_{ipt-1}^2 \left(\frac{\theta_0}{\sigma_0^2} + \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}} \log \theta_{ip}}{\sigma_\varepsilon^2} \right) \right. \\ &\quad \left. + \frac{1}{2} (\hat{\sigma}_{ipt-1}^2)^2 \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} + \frac{1}{2} \frac{\hat{\sigma}_{ipt-1}^2 + \sigma_\varepsilon^2}{\eta} \right]. \end{aligned}$$

Let $A \equiv \frac{1}{\sigma_0^2}$ and $B \equiv \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2}$. Then

$$\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta = \exp \left[\frac{A\theta_0 + B \log \theta_{ip}}{A+B} + \frac{B}{2(A+B)^2} + \frac{1}{2\eta} \left(\frac{1}{A+B} + \sigma_\varepsilon^2 \right) \right].$$

As $\psi_{it} \rightarrow \infty$ for all t , signals become perfectly informative and

$$\left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta \rightarrow \theta_{ip},$$

even with a finite sample.

To characterize wedges, define the threshold

$$\theta_{ip}^* = \exp \left[\theta_0 + \frac{1}{2A} \left(\frac{B}{A+B} + \frac{1}{\eta} + \frac{A+B}{\eta} \sigma_\varepsilon^2 \right) \right].$$

Then, $\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta > \theta_{ip}$ if $\theta_{ip} < \theta_{ip}^*$, and $\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta < \theta_{ip}$ if $\theta_{ip} > \theta_{ip}^*$. Moreover, since $\frac{\partial \log \theta_{ip}^*}{\partial B} > 0$, firms with lower ψ_{it} have a lower threshold. Thus, low- ψ firms underestimate $\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta$ for a wider range of high θ_{ip} .

Define the wedge

$$1 - \omega_{ipt} \equiv \frac{[\mathcal{B}_{it-1}]^\eta}{\theta_{ip}}.$$

Then output can be written as

$$y_{ipt}^* = (1 - \omega_{ipt}) \theta_{ip} z(\varphi_i)^\eta \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt}^\eta C_{pt} w_t^{-\eta}.$$

This implies that low- ψ firms underproduce high- θ_{ip} products and overproduce low- θ_{ip} products relative to the allocation implied by the true θ_{ip} . $\square$

Proof of Proposition 4, Learning Frictions and Product Switching

Proof. The firm's problem can be characterized by three cases:

$$V_{i1}(\hat{\theta}_{ip1}, 1) = \begin{cases} -f^s + \beta \left[x(\Xi_{i2}\mathcal{B}_{i1}(\hat{\theta}^0, 0)^\eta - \mathcal{F}^o) \right. \\ \quad \left. + (1-x)(\Xi_{i2}\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)^\eta - \mathcal{F}^o) \right] & , \text{ if } \mathcal{B}_{i1}^{ss} \geq \mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) > \mathcal{B}_{i1}^d \\ -f^s + \beta x \left[\Xi_{i2}\mathcal{B}_{i1}(\hat{\theta}^0, 0)^\eta - \mathcal{F}^o \right] & , \text{ if } \mathcal{B}_{i1}^d \geq \mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) \\ \beta \left[\Xi_{i2}\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)^\eta - \mathcal{F}^o \right] & , \text{ if } \mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) > \mathcal{B}_{i1}^{ss}, \end{cases}$$

where $\Xi_{i2} \equiv \frac{(\eta-1)^{\eta-1}}{\eta^\eta} \left(\frac{z(\varphi_i)}{w_2} \right)^{\eta-1} P_{p2}^\eta C_{p2}$. The first two cases correspond to search, while the last corresponds to no search.

When $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) = \mathcal{B}_{i1}^{ss}$, the firm is indifferent between searching and not searching. Thus,

$$(\mathcal{B}_{i1}^{ss})^\eta = -\frac{\mathcal{F}^{ss}}{\beta x \Xi_{i2}} + \left(\mathcal{B}(\hat{\theta}^0, 0) \right)^\eta. \quad (\text{A.47})$$

Since Ξ_{i2} is increasing in productivity $z(\varphi_i)$, the search threshold $\mathcal{B}_{i1}^{ss}$ is increasing in $z(\varphi_i)$.

When $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) = \mathcal{B}_{i1}^d$, the firm is indifferent between dropping and not retaining the product. Thus,

$$\mathcal{B}_{i1}^d = \left(\frac{\mathcal{F}^o}{\Xi_{i2}} \right)^{\frac{1}{\eta}}, \quad (\text{A.48})$$

which is a decreasing in productivity $z(\varphi_i)$.

Since

$$\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) = \exp \left\{ \frac{\hat{\theta}_{ip1}}{\eta} + \frac{1}{2} \left(\frac{\hat{\sigma}_{ip1}^2 + \sigma_\varepsilon^2}{\eta^2} \right) \right\},$$

where

$$\hat{\theta}_{ip1} = \frac{\left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip1} + \frac{\psi_{i1} \log s_{ip1}}{\sigma_\varepsilon^2} \right)}{\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} + \frac{\psi_{i1}}{\sigma_\varepsilon^2} \right)}, \text{ and } \hat{\sigma}_{ip1}^2 = \frac{1}{\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} + \frac{\psi_{i1}}{\sigma_\varepsilon^2} \right)},$$

Substituting the expressions for $\hat{\theta}_{ip1}$ and $\hat{\sigma}_{ip1}^2$, we can rewrite

$$\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) = \exp \left\{ \frac{1}{\eta} \left[\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} + \frac{\psi_{i1}}{\sigma_\varepsilon^2} \right)^{-1} \left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip1} + \frac{\psi_{i1} \log s_{ip1}}{\sigma_\varepsilon^2} + \frac{1}{2\eta} \right) + \frac{1}{2\eta} \sigma_\varepsilon^2 \right] \right\}.$$

Focusing on the terms involving ψ_{i1} , the derivative with respect to ψ_{i1} is proportional to

$$\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \right) \log s_{ip1} - \left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip1} + \frac{1}{2\eta} \right),$$

where the proportionality factor is strictly positive. Since $\log \theta_{ipt} = \log \theta_{ip} + \varepsilon_{ipt}$, where $\varepsilon_{ipt} \sim N(0, \sigma_\varepsilon^2)$, and $\log s_{ipt} = \log \theta_{ip} + v_{ipt}$, where $v_{ipt} \sim N(0, \frac{\sigma_\varepsilon^2}{\psi_{it}})$, taking expectations with respect to the noise terms yields

$$\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \right) \log \theta_{ip} - \left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip} + \frac{1}{2\eta} \right).$$

Therefore, for $\log \theta_{ip} > \theta_0 + \frac{\sigma_0^2}{2\eta}$, $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$ is increasing in ψ_{i1} , whereas for $\log \theta_{ip} < \theta_0 + \frac{\sigma_0^2}{2\eta}$, it is decreasing in ψ_{i1} . This implies that, for a given realization of θ_{ip1} , tenure $\tau = 1$, and productivity, firms with lower ψ_{i1} are more likely to underestimate $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$ when θ_{ip} is high. As a result, such firms are more likely to search for a new product ($\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) < \mathcal{B}_{i1}^{ss}$) or even drop the product ($\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) < \mathcal{B}_{i1}^d$), even though the true θ_{ip} is high enough to justify continued operation without search.

Conversely, when θ_{ip} is low, firms with lower ψ_{i1} are more likely to overestimate $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$. In this case, they are more likely to continue operating the product, even though the true θ_{ip} would justify dropping it.

Furthermore, the likelihood that firms operate misaligned product lines for a given ψ_i gets higher if the firm has higher $z(\varphi_i)$. This is because, given all else equal, higher $z(\varphi_i)$ increases the search threshold $\mathcal{B}_{i1}^{ss}$ and decreases the drop threshold $\mathcal{B}_{i1}^d$ decreases in $z(\varphi_i)$, which widens the range of posterior mean $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$, over which firms both continue operating and search. This expanded region induces more firms—even those with relatively precise beliefs (high ψ_i)—to switch away from high- θ_{ip} products or retain

low- θ_{ip} products. As a result, misallocation driven by learning frictions becomes more pronounced among high-productivity firms. $\square$

B Empiric

B.1 Full Tables: Product Dropping

Table A1: Probability of Product Drop (PV)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
$\log(PV_{ipt})$	-0.055*** (0.002)	-0.054*** (0.002)	-0.053*** (0.002)	-0.053*** (0.002)	-0.052*** (0.002)
$\log(\text{Emp})$			-0.020*** (0.002)	-0.009*** (0.002)	-0.012*** (0.002)
$\log(\text{Total Assets})$			0.003*** (0.001)	0.019*** (0.004)	0.019*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
# products			0.007*** (0.000)	0.007*** (0.000)	0.006*** (0.000)
$\log(\text{CA})$			-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)
$\log(\text{CS})$			-0.004*** (0.001)	-0.004*** (0.001)	-0.003*** (0.001)
TFP			-0.004* (0.002)	-0.006** (0.002)	-0.007*** (0.003)
$\log(\text{Prod. Emp})$				-0.009*** (0.001)	-0.010*** (0.001)
$\log(\text{Equipment})$				-0.028*** (0.004)	-0.029*** (0.004)
$\log(\text{Structure})$				0.011*** (0.003)	0.011*** (0.003)
$\log(\text{Material})$				-0.004*** (0.001)	-0.004*** (0.001)
$\log(\text{Energy})$				0.004*** (0.001)	0.003*** (0.001)
Tenure		0.004*** (0.000)			0.004*** (0.001)
Tenure squared		-0.000*** (0.000)			-0.000*** (0.000)
Constant	0.943*** (0.010)	0.942*** (0.010)	0.965*** (0.015)	0.978*** (0.016)	1.004*** (0.017)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Probability of Product Drop (θ_{ip})

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-1.212*** (0.029)	-1.028*** (0.022)	-1.189*** (0.029)	-1.187*** (0.029)	-1.074*** (0.021)
log(Emp)			-0.035*** (0.002)	-0.015*** (0.002)	-0.016*** (0.002)
log(Total Assets)			0.002* (0.001)	0.023*** (0.004)	0.021*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
# products			0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
log(CA)			0 (0.001)	0 (0.001)	0.002*** (0.001)
log(CS)			-0.005*** (0.001)	-0.005*** (0.001)	-0.002*** (0.001)
TFP			-0.011*** (0.002)	-0.020*** (0.002)	-0.013*** (0.002)
log(Prod. Emp)				-0.009*** (0.001)	-0.010*** (0.001)
log(Equipment)				-0.025*** (0.004)	-0.025*** (0.004)
log(Structure)				0.005** (0.003)	0.006** (0.003)
log(Material)				-0.013*** (0.001)	-0.008*** (0.001)
log(Energy)				0.001 (0.001)	0.002* (0.001)
log(PV_{ipt})		-0.021*** (0.001)			-0.017*** (0.001)
Tenure		0.041*** (0.001)			0.042*** (0.001)
Tenure squared		-0.001*** (0.000)			-0.001*** (0.000)
Constant	1.053*** (0.015)	0.963*** (0.014)	1.164*** (0.021)	1.221*** (0.022)	1.111*** (0.022)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A3: Probability of Product Drop (Young Firm)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
Young Firm _{it}	-0.084*** (0.004)	-0.073*** (0.003)	-0.092*** (0.004)	-0.093*** (0.004)	-0.075*** (0.003)
log(Emp)			-0.050*** (0.002)	-0.025*** (0.002)	-0.015*** (0.002)
log(Total Assets)			-0.008*** (0.001)	0.015*** (0.005)	0.019*** (0.004)
# establishments			-0.004*** (0.000)	-0.004*** (0.000)	-0.002*** (0.000)
# products			0.011*** (0.001)	0.011*** (0.001)	0.006*** (0.000)
log(CA)			-0.003*** (0.001)	-0.002*** (0.001)	-0.001** (0.001)
log(CS)			-0.002** (0.001)	-0.002** (0.001)	-0.003*** (0.001)
TFP			-0.014*** (0.002)	-0.027*** (0.003)	-0.008*** (0.003)
log(Prod. Emp)				-0.008*** (0.001)	-0.010*** (0.001)
log(Equipment)				-0.032*** (0.004)	-0.029*** (0.004)
log(Structure)				0.011*** (0.003)	0.011*** (0.003)
log(Material)				-0.019*** (0.001)	-0.005*** (0.001)
log(Energy)				-0.002* (0.001)	0.003*** (0.001)
log(PV _{ipt})		-0.054*** (0.002)			-0.052*** (0.002)
Tenure		0.002*** (0.000)			0.003*** (0.000)
Tenure squared		-0.000*** (0.000)			-0.000*** (0.000)
Constant	0.591*** (0.000)	0.951*** (0.011)	0.828*** (0.015)	0.918*** (0.016)	1.038*** (0.018)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A4: Probability of Product Drop ($PV \times \text{Young}$)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
$\log(PV_{ipt})$	-0.056*** (0.002)	-0.055*** (0.002)	-0.054*** (0.002)	-0.054*** (0.002)	-0.052*** (0.002)
Young Firm _{it}	-0.163*** (0.008)	-0.142*** (0.007)	-0.159*** (0.008)	-0.160*** (0.008)	-0.141*** (0.007)
$\log(PV_{ipt}) \times \text{Young}_{it}$	0.012*** (0.001)	0.011*** (0.001)	0.011*** (0.001)	0.011*** (0.001)	0.010*** (0.001)
$\log(\text{Emp})$			-0.025*** (0.002)	-0.013*** (0.002)	-0.015*** (0.002)
$\log(\text{Total Assets})$			0.001 (0.001)	0.019*** (0.004)	0.019*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
# products			0.007*** (0.000)	0.007*** (0.000)	0.006*** (0.000)
$\log(\text{CA})$			-0.002*** (0.001)	-0.001* (0.001)	-0.001* (0.001)
$\log(\text{CS})$			-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)
TFP			-0.005** (0.002)	-0.008*** (0.003)	-0.009*** (0.003)
$\log(\text{Prod. Emp})$				-0.009*** (0.001)	-0.010*** (0.001)
$\log(\text{Equipment})$				-0.029*** (0.004)	-0.029*** (0.004)
$\log(\text{Structure})$				0.011*** (0.003)	0.011*** (0.003)
$\log(\text{Material})$				-0.004*** (0.001)	-0.005*** (0.001)
$\log(\text{Energy})$				0.004*** (0.001)	0.003*** (0.001)
Tenure		0.003*** (0.000)			0.003*** (0.000)
Tenure squared		-0.000*** (0.000)			-0.000*** (0.000)
Constant	0.960*** (0.011)	0.957*** (0.011)	1.012*** (0.016)	1.027*** (0.017)	1.044*** (0.018)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A5: Probability of Product Drop ($\theta_{ip} \times \text{Young}$)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-1.221*** (0.030)	-1.032*** (0.022)	-1.196*** (0.030)	-1.194*** (0.030)	-1.081*** (0.022)
Young Firm $_{it}$	-0.411*** (0.022)	-0.371*** (0.023)	-0.402*** (0.022)	-0.403*** (0.022)	-0.368*** (0.022)
$\theta_{ip} \times \text{Young}_{it}$	0.395*** (0.044)	0.514*** (0.043)	0.367*** (0.043)	0.368*** (0.043)	0.497*** (0.043)
log(Emp)			-0.040*** (0.002)	-0.020*** (0.002)	-0.018*** (0.002)
log(Total Assets)			0 (0.001)	0.022*** (0.004)	0.020*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.001*** (0.000)
# products			0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
log(CA)			0 (0.001)	0.001* (0.001)	0.002*** (0.001)
log(CS)			-0.004*** (0.001)	-0.004*** (0.001)	-0.002*** (0.001)
TFP			-0.012*** (0.002)	-0.021*** (0.002)	-0.014*** (0.002)
log(Prod. Emp)				-0.009*** (0.001)	-0.010*** (0.001)
log(Equipment)				-0.026*** (0.004)	-0.026*** (0.004)
log(Structure)				0.005* (0.003)	0.006** (0.003)
log(Material)				-0.014*** (0.001)	-0.008*** (0.001)
log(Energy)				0 (0.001)	0.002* (0.001)
log(PV_{ipt})		-0.021*** (0.001)			-0.017*** (0.001)
Tenure		0.039*** (0.001)			0.039*** (0.001)
Tenure squared		-0.001*** (0.000)			-0.001*** (0.000)
Constant	1.071*** (0.015)	0.980*** (0.014)	1.222*** (0.021)	1.280*** (0.023)	1.146*** (0.022)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A6: Probability of Product Drop ($\theta_{ip} \times$ Management)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
Management	0.356*** (0.040)	0.465*** (0.041)	0.341*** (0.033)	0.406*** (0.033)	0.348*** (0.035)	0.427*** (0.036)
$\theta_{ip} \times$ MOPS	-0.983*** (0.080)	-1.036*** (0.082)	-0.768*** (0.061)	-0.832*** (0.062)	-0.837*** (0.066)	-0.912*** (0.068)
θ_{ip}	-1.211*** (0.045)	-1.055*** (0.041)	-1.292*** (0.037)	-1.127*** (0.033)	-1.289*** (0.038)	-1.119*** (0.035)
log(Emp)		-0.015*** (0.002)		-0.015*** (0.002)		-0.015*** (0.002)
log(Total Assets)		0.021*** (0.004)		0.020*** (0.004)		0.020*** (0.004)
# establishments		-0.001*** (0.000)		-0.002*** (0.000)		-0.002*** (0.000)
# products		0.005*** (0.000)		0.005*** (0.000)		0.005*** (0.000)
log(CA)		0.002*** (0.001)		0.002*** (0.001)		0.002*** (0.001)
log(CS)		-0.001 (0.001)		-0.001 (0.001)		-0.001 (0.001)
TFP		-0.012*** (0.002)		-0.012*** (0.002)		-0.012*** (0.002)
log(Prod. Emp)		-0.011*** (0.001)		-0.011*** (0.001)		-0.011*** (0.001)
log(Equipment)		-0.025*** (0.004)		-0.025*** (0.004)		-0.025*** (0.004)
log(Structure)		0.005** (0.003)		0.005** (0.003)		0.005** (0.003)
log(Material)		-0.007*** (0.001)		-0.007*** (0.001)		-0.007*** (0.001)
log(Energy)		0.002*** (0.001)		0.002*** (0.001)		0.002*** (0.001)
log(PV_{ipt})		-0.017*** (0.001)		-0.017*** (0.001)		-0.017*** (0.001)
Tenure		0.041*** (0.001)		0.041*** (0.001)		0.041*** (0.001)
Tenure squared		-0.001*** (0.000)		-0.001*** (0.000)		-0.001*** (0.000)
Constant	1.014*** (0.022)	1.048*** (0.028)	0.998*** (0.019)	1.056*** (0.025)	1.012*** (0.019)	1.064*** (0.026)

Notes: Managerial ability is measured using the Management and Organizational Practices Survey (MOPS). Columns (1)–(2) use the overall management index as in Bloom et al. (2019), columns (3)–(4) use the data-availability measure (Q27), and columns (5)–(6) use the data use intensity measure (Q28). Estimates for firm and product-year fixed effects are suppressed. Standard errors clustered at the SIC5 level are reported in parentheses. Observations are unweighted. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

B.2 Full Tables: Firm-Level Evidence

Table A7: Product Add and Retention (Young Firm)

	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$
Young Firm _{it}	0.140*** (0.005)	0.148*** (0.005)	0.148*** (0.005)
log(Emp)		0.032*** (0.003)	0.029*** (0.002)
log(Total Assets)		0.010*** (0.001)	-0.005 (0.004)
# establishments		0.012*** (0.004)	0.012*** (0.003)
# products		-0.027*** (0.002)	-0.027*** (0.002)
log(CA)		-0.003*** (0.001)	-0.003** (0.001)
log(CS)		-0.006*** (0.001)	-0.007*** (0.001)
TFP		-0.001 (0.003)	0.008*** (0.003)
log(Prod. Emp)			-0.008*** (0.001)
log(Equipment)			0.021*** (0.003)
log(Structure)			-0.008*** (0.003)
log(Material)			0.010*** (0.001)
log(Energy)			0.006*** (0.001)
Constant	0.350*** (0.002)	0.220*** (0.015)	0.174*** (0.018)

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A8: High-Performing Product Add and Retention (Young Firm)

	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$
Young Firm _{it}	-0.115*** (0.016)	-0.114*** (0.017)	-0.112*** (0.017)
log(Emp)		-0.018** (0.008)	-0.018** (0.009)
log(Total Assets)		-0.007 (0.005)	0.037*** (0.012)
# establishments		-0.007*** (0.003)	-0.007*** (0.003)
# products		0.020*** (0.003)	0.020*** (0.003)
log(CA)		0.003 (0.003)	0.002 (0.003)
log(CS)		-0.004 (0.003)	-0.003 (0.003)
TFP		-0.020** (0.008)	-0.034*** (0.009)
log(Prod. Emp)			0.021*** (0.003)
log(Equipment)			-0.025*** (0.009)
log(Structure)			-0.015* (0.009)
log(Material)			-0.016*** (0.004)
log(Energy)			-0.013*** (0.005)
Constant	0.347*** (0.003)	0.463*** (0.041)	0.516*** (0.049)

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table A9: Product Add/Drop and Firm Performance

	ΔV_{add}	ΔEmp	ΔTFP	ΔLP
$Add_{t:t+5}$	0.830*** (0.019)	0.802*** (0.019)	0.735*** (0.019)	0.762*** (0.019)
$Drop_{t:t+5}$	-0.754*** (0.019)	-0.731*** (0.018)	-0.686*** (0.017)	-0.713*** (0.018)
log(Emp)	-0.187*** (0.009)	-0.604*** (0.008)	0.099*** (0.007)	0.200*** (0.007)
log(Total Assets)	-0.071*** (0.006)	-0.021*** (0.006)	-0.026** (0.012)	-0.018 (0.011)
# establishments	-0.027*** (0.008)	-0.024*** (0.007)	-0.026*** (0.008)	-0.028*** (0.008)
# products	0.077*** (0.006)	0.079*** (0.005)	0.070*** (0.005)	0.072*** (0.006)
log(CA)	-0.003* (0.002)	0.006*** (0.002)	0.002 (0.002)	0.003 (0.002)
log(CS)	0.014*** (0.002)	0.014*** (0.002)	0.011*** (0.002)	0.012*** (0.002)
TFP	-0.561*** (0.012)	0.130*** (0.006)	-0.369*** (0.022)	-0.088*** (0.026)
log(Prod. Emp)	0.021*** (0.003)	0.028*** (0.003)	0.019*** (0.003)	0.001 (0.003)
log(Equipment)	0.030*** (0.005)	0.075*** (0.005)	0.059*** (0.007)	0.029*** (0.006)
log(Structure)	-0.046*** (0.005)	-0.028*** (0.005)	-0.028*** (0.008)	-0.035*** (0.007)
log(Material)	-0.049*** (0.006)	0.068*** (0.003)	0.023*** (0.006)	-0.042*** (0.008)
log(Energy)	-0.030*** (0.004)	0.042*** (0.002)	0.010*** (0.003)	0.035*** (0.004)
Constant	2.252*** (0.052)	0.424*** (0.047)	-0.415*** (0.081)	-0.633*** (0.100)
Observations	682,000	682,000	648,000	648,000
Fixed effects	i, jt	i, jt	i, jt	i, jt
Controls	Full	Full	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

B.3 Robustness

Table A10: Probability of Product Drop (Alternative θ_{ip} Estimates)

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
θ_{ip}	0.037 (0.037)	-0.013 (0.037)	-0.608*** (0.016)	-0.617*** (0.016)	-0.491*** (0.015)	-0.495*** (0.015)	-0.313*** (0.016)	-0.318*** (0.016)
Young Firm		-0.453*** (0.025)		-0.344*** (0.016)		-0.303*** (0.017)		-0.130*** (0.020)
$\theta_{ip} \times \text{Young}$		0.439*** (0.042)		0.376*** (0.025)		0.287*** (0.026)		0.083*** (0.030)
Constant	0.781*** (0.021)	0.849*** (0.020)	0.946*** (0.021)	0.983*** (0.021)	0.897*** (0.019)	0.935*** (0.019)	0.939*** (0.016)	0.974*** (0.017)
Observations	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000
Controls	Full	Full	Full	Full	Full	Full	Full	Full
θ_{ip} estimation	2obs	2obs	5yr rolling	5yr rolling	10yr rolling	10yr rolling	15yr rolling	15yr rolling
VARIABLES	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
θ_{ip}	-0.050*** (0.001)	-0.050*** (0.001)	-0.158*** (0.005)	-0.160*** (0.005)	-0.049*** (0.001)	-0.050*** (0.001)	-0.160*** (0.005)	-0.163*** (0.005)
Young Firm		-0.108*** (0.004)		-0.148*** (0.005)		-0.108*** (0.004)		-0.148*** (0.005)
$\theta_{ip} \times \text{Young}$		0.025*** (0.002)		0.076*** (0.007)		0.027*** (0.002)		0.076*** (0.007)
Constant	0.553*** (0.020)	0.584*** (0.020)	0.646*** (0.020)	0.678*** (0.020)	0.555*** (0.021)	0.585*** (0.020)	0.646*** (0.020)	0.677*** (0.020)
Observations	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000
Adjusted R^2	0.495	0.497	0.494	0.496	0.495	0.497	0.494	0.496
Controls	Full	Full	Full	Full	Full	Full	Full	Full
θ_{ip} estimation	Norm (jt)	Norm (jt)	Percentile (jt)	Percentile (jt)	Norm (pt)	Norm (pt)	Percentile (pt)	Percentile (pt)

Notes: Columns are based on alternative specification of estimating θ_{ip} . Columns (1)-(2) are based on the first 2 observations for each product-firm pair, (3)-(4) are estimates with 5 year rolling-year basis, (5)-(6) are with 10 year rolling-year basis, (7)-(8) are with 15 year rolling-year basis, (9)-(10) are based on standardization with the industry-year distribution, (11)-(12) are based on the percentile rankings within industry-year, (13)-(14) are based on standardization with the product-year distribution, and (15)-(16) are based on the percentile rankings within product-year. Full-control specifications include the same set of controls as in the baseline regressions. Estimates for product-year and firm fixed effects are suppressed. Observations are unweighted. Standard errors clustered at the SIC5 level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Probability of Product Drop (Alternative θ_{ip} Estimates)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-0.563*** (0.016)	-0.580*** (0.017)
Young Firm		-0.317*** (0.014)
$\theta_{ip} \times \text{Young}$		0.394*** (0.027)
Constant	0.855*** (0.024)	0.892*** (0.024)
Observations	1,622,000	1,622,000
Adjusted R^2	0.493	0.495
Controls	Full	Full
θ_{ip} estimation	No controls	No controls

Notes: Results are based on alternative specification of estimating θ_{ip} without firm controls. Estimates for product-year and firm fixed effects are suppressed. Observations are unweighted. Standard errors clustered at the SIC5 level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Probability of Product Drop (θ_{ip} , bootstrapped standard errors)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-1.212*** (0.094)	-1.028*** (0.119)	-1.189*** (0.095)	-1.187*** (0.095)	-1.074*** (0.120)
log(Emp)			-0.035*** (0.001)	-0.015*** (0.001)	-0.016*** (0.001)
log(Total Assets)			0.002*** (0.000)	0.023*** (0.001)	0.021*** (0.001)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
# products			0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
log(CA)			0 (0.000)	0 (0.000)	0.002*** (0.000)
log(CS)			-0.005*** (0.000)	-0.005*** (0.000)	-0.002*** (0.000)
TFP			-0.011*** (0.001)	-0.020*** (0.001)	-0.013*** (0.001)
log(Prod. Emp)				-0.009*** (0.001)	-0.010*** (0.000)
log(Equipment)				-0.025*** (0.001)	-0.025*** (0.001)
log(Structure)				0.005*** (0.001)	0.006*** (0.001)
log(Material)				-0.013*** (0.001)	-0.008*** (0.001)
log(Energy)				0.001 (0.001)	0.002*** (0.001)
log(PV_{ipt})		-0.021*** (0.002)			-0.017*** (0.002)
Tenure		0.041*** (0.000)			0.042*** (0.000)
Tenure squared		-0.001*** (0.000)			-0.001*** (0.000)

Notes: Estimates for product-year and firm fixed effects are suppressed. Observations are unweighted. Bootstrapped standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Probability of Product Drop ($\theta_{ip} \times \text{Young}$, bootstrapped standard errors)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-1.221*** (0.091)	-1.032*** (0.118)	-1.196*** (0.092)	-1.194*** (0.092)	-1.081*** (0.120)
Young Firm _{it}	-0.411*** (0.055)	-0.371*** (0.042)	-0.402*** (0.056)	-0.403*** (0.056)	-0.368*** (0.043)
$\theta_{ip} \times \text{Young}_{it}$	0.395*** (0.096)	0.514*** (0.072)	0.367*** (0.098)	0.368*** (0.099)	0.497*** (0.073)
log(Emp)			-0.040*** (0.001)	-0.020*** (0.001)	-0.018*** (0.001)
log(Total Assets)			0 (0.000)	0.022*** (0.001)	0.020*** (0.001)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.001*** (0.000)
# products			0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
log(CA)			0 (0.000)	0.001** (0.000)	0.002*** (0.000)
log(CS)			-0.004*** (0.000)	-0.004*** (0.000)	-0.002*** (0.000)
TFP			-0.012*** (0.001)	-0.021*** (0.001)	-0.014*** (0.001)
log(Prod. Emp)				-0.009*** (0.001)	-0.010*** (0.000)
log(Equipment)				-0.026*** (0.001)	-0.026*** (0.001)
log(Structure)				0.005*** (0.001)	0.006*** (0.001)
log(Material)				-0.014*** (0.001)	-0.008*** (0.001)
log(Energy)				0 (0.000)	0.002*** (0.001)
log(PV_{ipt})		-0.021*** (0.002)			-0.017*** (0.002)
Tenure		0.039*** (0.000)			0.039*** (0.000)
Tenure squared		-0.001*** (0.000)			-0.001*** (0.000)

Notes: Estimates for product-year and firm fixed effects are suppressed. Observations are unweighted. Bootstrapped standard errors are reported in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

Table A14: Probability of Product Drop ($\theta_{ip} \times \text{Firm Age}$)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-0.816*** (0.038)	-0.628*** (0.033)
$\theta_{ip} \times \text{Firm Age}_{it}$	-0.022*** (0.002)	-0.024*** (0.002)
log(Emp)		-0.015*** (0.002)
log(Total Assets)		0.021*** (0.004)
# establishments		-0.002*** (0.000)
# products		0.005*** (0.000)
log(CA)		0.001** (0.001)
log(CS)		-0.004*** (0.001)
TFP		-0.013*** (0.002)
log(Prod. Emp)		-0.010*** (0.001)
log(Equipment)		-0.025*** (0.004)
log(Structure)		0.006** (0.003)
log(Material)		-0.008*** (0.001)
log(Energy)		0.002* (0.001)
log(PV_{ipt})		-0.018*** (0.001)
Tenure		0.041*** (0.001)
Tenure squared		-0.001*** (0.000)
Constant	1.018*** (0.014)	1.070*** (0.022)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Firm Age is measured as the age of the firm in years. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A15: Number of Input Materials Used per Product by Firms

	Including entry and exit			Continuers only		
	all	age 0-5	age 6 +	all	age 0-5	age 6 +
Mean	20.30	19.45	20.43	21.33	21.50	21.31
Stdev.	(20.29)	(19.86)	(20.35)	(21.00)	(21.34)	(20.95)

Notes: Firms in the manufacturing sector from 1982 to 2007.

B.4 Product-Firm Proximity

Up to this point, we have focused on the relationship between product dropping and adding, as well as the relative match quality of products. We now turn to examining the proximity between newly added products and a firm's existing product portfolio. The literature finds that firms tend to produce products close to each other in terms of various measures, such as industry classification or the composition of material inputs used (e.g., [Bernard et al., 2010](#); [Boehm et al., 2022](#)). Nonetheless, we also observe many instances in which firms expand into entirely new industries.

To examine whether the proximity of added products to a firm's existing product portfolio varies over the firm life-cycle, we use the share of input materials used to produce a newly added product that are also used for the firm's existing products as a proxy for product proximity.^{44,45} Let $MaterialCloseness_{ijt}^{add}$ denote the average of the proximity measure of products added by firm i between t and $t + 5$. The mean (standard deviation) of the material-based closeness of added products in the churning case is 0.708 (0.271). Table A15 shows the number of input materials used per product by firms across age groups. Focusing on the continuing firms, we find that young firms use slightly more variety of input materials to produce each product, and the variance is also larger compared to mature firms.

⁴⁴The CMF provides establishment-level data on the list of material inputs used and associated expenditures. By focusing on single-product establishments, we identify the set of material inputs used in the production of each product.

⁴⁵We are in the process of exploring an expenditure-weighted version of the material input-based proximity measure, following [Boehm et al. \(2022\)](#).

We then estimate the following regression model:

$$MaterialCloseness_{ijt}^{add} = \alpha + \beta \mathcal{I}_{ijt}^{young} + X_{ijt}\gamma + \delta_i + \delta_{jt} + \varepsilon_{ijt}. \quad (\text{B.49})$$

Table A16 shows that relative to mature firms, young firms tend to add and retain products that are more distant from their existing product portfolio. Results are also consistent with industry-based proximity.

Table A16: Material-Based Closeness of Added Products (Young Firm)

	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$
Young Firm _{it}	-0.005 (0.005)	-0.003 (0.004)	-0.003 (0.004)
log(Emp)		-0.012*** (0.002)	-0.003 (0.003)
log(Total Assets)		-0.004*** (0.001)	-0.001 (0.004)
# establishments		0.005*** (0.002)	0.005*** (0.002)
# products		-0.024*** (0.002)	-0.024*** (0.002)
log(CA)		-0.003*** (0.001)	-0.003*** (0.001)
log(CS)		-0.007*** (0.001)	-0.007*** (0.001)
TFP		-0.003 (0.002)	-0.007*** (0.002)
# materials		0.005*** (0.000)	0.005*** (0.000)
log(Prod. Emp)			-0.001 (0.001)
log(Equipment)			-0.005* (0.003)
log(Structure)			0.004* (0.002)
log(Material)			-0.005*** (0.001)
log(Energy)			-0.007*** (0.001)
Constant	0.520*** (0.003)	0.489*** (0.011)	0.521*** (0.013)
Observations	198,000	198,000	198,000
Fixed effects	<i>i, jt</i>	<i>i, jt</i>	<i>i, jt</i>

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

B.5 Discussion on Product-Firm Expertise Estimation

One alternation of the expertise estimation is to use product-year and firm-year fixed effects as follows, instead of controlling for the time-varying firm and industry characteristics:

$$y_{ipjt} = \theta_{ip} + X_{ipjt}\gamma + \delta_{pt} + \delta_{it} + \varepsilon_{ipjt}, \quad (\text{B.50})$$

where δ_{pt} is a product-year fixed effect, and δ_{it} is a firm-year fixed effect.

δ_{pt} controls for product specific characteristics as well as product-year specific shocks. Again, this is because each product is subject to different production technology or demand structure. Therefore, it is possible that there are several other effects affecting the value of shipment, attributed to product-year specific technology or demand shocks.

In a similar fashion, firms or entrepreneurs have different base sets of available resources or ability to manufacture products (i.e. a financing constraint, customer capital, brand values, and marketing resources, etc.). The value of shipment for each product can also be influenced by these firm-level properties. Furthermore, any types of industry-specific characteristics that can give a substantial impact on the total value of shipment are absorbed by the firm-year fixed effect. Therefore, equation (B.50) could be one alternative of the baseline quality estimation in (4.38).

However, estimation based on (B.50) faces several limitations due to the inclusion of multiple fixed effects. In particular, estimating firm-product and firm-year fixed effects simultaneously leads to the exclusion of a non-negligible portion of the sample, as some products appear in only one firm or some firms produce only a single product in a given year. Therefore, an alternative approach is to replace the firm-year fixed effect δ_{it} with an extensive set of firm-level controls X_{ijt} , which is our baseline specification (4.38).

Alternatively, we could implement an AKM-style specification (Abowd et al., 1999) and estimate the firm-product match quality by taking the time average of the residuals.

These specifications include

$$y_{ipjt} = X_{ipjt}\gamma + \delta_{pt} + \delta_{it} + \varepsilon_{ipjt} \quad (\text{B.51})$$

$$y_{ipjt} = X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + X_{pt}\gamma_3 + \delta_p + \delta_i + \delta_t + \varepsilon_{ipjt} \quad (\text{B.52})$$

$$y_{ipjt} = X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_{pt} + \delta_i + \delta_t + \varepsilon_{ipjt}. \quad (\text{B.53})$$

A critical limitation of this approach is the lack of clarity regarding what the time-varying residuals capture, and whether their average offers a more accurate measure of firm-product match quality than the baseline model.